Supplemental Material for "Emergent $\mathcal{PT}$ Symmetry and Exceptional Points in a Driven Dirac Impurity"

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This Supplemental Material gives the derivations, certificates, and validation checks that support the main article. Throughout, $\mathcal{PT}$ denotes parity–time symmetry.

NOTATION AND PARAMETER CONVENTIONS

$t_{\rm av}, \tau, T_{\rm b}$**:** Wigner center time $(t+t')/2$, relative time $t-t'$, and reservoir temperature.

η, ζ_σ**:** Chiral bath index $\eta = \pm$ and spin sign $\zeta_\uparrow = +1$, $\zeta_\downarrow = -1$.

$\lambda, j_\alpha, \chi_K$**:**
SOC strength, eigenvalues of the bath-density-normalized Schrieffer–Wolff matrix $\rho_b^{1/2} J^{\rm SW} \rho_b^{1/2}$, and the one-loop coefficient defined by $dj/d\ell = \chi_K j^2$. For an antiferromagnetic Hermitian reference, $j_{\rm scr}$ denotes the largest positive eigenvalue.

$\alpha_{\rm b}, k_{\max}, D_{\rm uv}$**:**
Quadratic bath-dispersion coefficient, radial momentum cutoff, and ultraviolet energy cutoff. The normalized $k=0$ core figures contain neither cutoff; $k_{\max} = \pi/4$ is used in the causal gate and $D_{\rm uv} = \pi/4$ only in the sharp-cutoff reachability audit.

$\epsilon_\xi, \epsilon_d, \tilde{\epsilon}_\xi$**:** Bare impurity gate in the benchmark notation, bare gate in the full-bath calculation, and renormalized level of the scalar slave-boson benchmark. The latter includes the real constraint and projected one-body shifts.

$W_k, W, \Lambda_\eta^{\rm PV}$**:**
Static bare impurity–bath hybridization and principal-value bath shift. The drive does not set W_k; at the saddle the vertex is rW_k.

$\beta_0, \tilde{\beta}$**:** Signed frozen drive-control amplitude and nonnegative slave-boson mean-field (SBMF) projection $\tilde{\beta} = |\beta_0||b_c|$. On the two paths described below the table, $G = |\beta_0|$ or $G = 0.20\beta_0$, so β_0 is not itself a decay rate.

z, z_W, s**:** Complex frequency, Wilson discretization shift, and frozen-core half-splitting ($E_+ - E_- = 2s$). The physical retarded boundary is $z = \mu + \omega + i0^+$ when ω is measured from μ; ζ_σ denotes the spin sign.

s_a, δ_a**:** Eigenvalues $\mathsf{s}_a = e^{2i\delta_a}$ and eigenphases of the complete unitary scattering matrix. They are distinct from the EP pole splitting s and from the fixed chiral labels $\eta = \pm$.

$D, G, \Gamma_{\rm PT}, \bar{\Gamma}$**:**
Constant-core coherent splitting, constant-core relative channel, projected relative decay imbalance, and common bath width. The symbols $\tilde{\Gamma}_\eta$ and $\Gamma_\sigma^{\rm net}$ denote, respectively, the chiral frozen-kernel width and the physical net channel width.

ω, Ω, ω_0**:** Fourier frequency, drive frequency, and a selected spectral pole or resonance frequency; none of these denotes temperature.

$\mathcal{E}_n, E_{\rm EP}, \Lambda(\omega)$**:**
Non-Hermitian eigenvalues, EP eigenvalue, and real bath self-energy.

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* JNCASR is the author's doctoral affiliation; this work was carried out independently

$\mathcal{P}, \mathfrak{a}, \mathfrak{b}$**:** Full-kernel Schur polynomial and the two local contact denominators; these are distinct from the bath coefficient $\alpha_{\rm b}$.

$\mathcal{F}_{\rm dist}, \mathcal{F}_{\rm MF}$**:**
Distribution matrix and mean-field free-energy functional.

R, Λ_f, D_d**:**
Finite-U Kotliar–Ruckenstein renormalization matrix and constraint field (App. N), and impurity double occupancy $\langle n_{d\uparrow} n_{d\downarrow}\rangle$. The constraint field is distinct from the Wilson parameter Λ and the real bath self-energy $\Lambda(\omega)$.

For clarity, the same constant two-component matrix is written in three conventions:

representation	constant matrix	identification
spin basis	$D\sigma_x + iG\sigma_z$	$D = \Delta_{\rm eff}$, $G = \Gamma_{\rm PT}$
fixed chiral basis	$D\sigma_z + iG\sigma_x$	Hadamard rotation
companion algebra	$\gamma\sigma_x + i\beta\sigma_z$	$\gamma = D$, $\beta = G$

The frozen drive coordinate is instead β_0; it must not be identified with the matrix coefficient G or with the companion symbol β. The normalized geometry plots use $D = 0.50$ and $G = |\beta_0|$, whereas the small-width causal/flow audit uses $D = 0.075 + 0.050\beta_0$ and $G = 0.20\beta_0$, giving $D = 0.10$ at the core point. The geometry path has crossings at $\beta_0 = \pm 0.50$; the small-width audit uses only $0.35 \le \beta_0 \le 0.55$ and crosses at $\beta_0 = 0.50$. These are different parameter paths, with neither identical coherent coefficients nor identical physical decay rates.

Appendix A: Impurity-block EP condition and condition-number scaling

This appendix records the impurity-block algebra used in the main text. The Hadamard rotation itself is shown in Eq. (22); here we keep only the EP and condition-number scaling that underlies the diagnostic interpretation.

The frozen impurity block is $h_d = \left[\begin{smallmatrix} \tilde\epsilon_\xi + D & iG \\ iG & \tilde\epsilon_\xi - D \end{smallmatrix}\right]$, with eigenvalues $E_\pm = \tilde\epsilon_\xi \pm s$ and splitting $s = \sqrt{D^2 - G^2}$. The regimes are $\mathcal{PT}$-unbroken ($D > G$), exceptional ($D = G$), and $\mathcal{PT}$-broken ($G > D$) for positive parameters. Right eigenvectors may be chosen as $|r_\pm\rangle = [G,\, i(D \mp s)]^T$, so $\det R = 2iGs, \qquad R = (|r_+\rangle, |r_-\rangle)$. Thus, near the bare EP,

$$\kappa_{\rm imp} = \|R\|\|R^{-1}\| \sim \frac{C}{|s|}, \tag{A1}$$

with a nonsingular constant C. The standard local U does not cut off this algebraic divergence on the equal-velocity, scalar-hybridization linearized submanifold because it preserves the complexified pseudospin symmetry. At fixed real frequency with a strictly positive decay margin, the complete retarded resolvent stays finite despite the singular eigenbasis; the analytically continued Green function still has its resonance poles. Curvature, unequal velocities, frequency dependence, or component-dependent vertices can lift the exact symmetry and must be analyzed separately.

1. Slave-boson scaling and the core EP

The bare impurity–bath hybridization W in Eq. (3) of the main article is time independent. At the saddle it becomes rW. Since the projected impurity self-energy is $\Sigma_d = W^\dagger G_c W$, every induced impurity channel carries the same factor r^2: $\Gamma_{\rm PT} = r^2\Gamma^{(0)}_{\rm PT}(\beta_0)$, $\Delta_{\rm eff} = r^2\Delta^{(0)}_0$ and $\bar\Gamma = r^2\Gamma_0$. Consequently

$$h^2_{\rm rel} = r^4 \left\{ [\Delta^{(0)}_0]^2 - [\Gamma^{(0)}_{\rm PT}(\beta_0)]^2 \right\} I. \tag{A2}$$

The common r^4 cancels from the local core condition for every $r > 0$. The cancellation requires one renormalization factor for both spins. A finite-U saddle has spin-dependent factors, $R \neq rI$, and a traceless constraint field; the projected kernel $R\,\Sigma_d R + \vec\Lambda_f \cdot \vec\sigma$ then does not factorize, and the EP moves (App. N). At $r = 0$ the result is an ordinary decoupled degeneracy. A bath-dressed finite-k minimum can drift because the SOC-resolved Schur terms do not remain a common scalar shift; the audited search reaches the scan boundary and is therefore not classified as an EP. At $k = 0$, by contrast, $\varepsilon_{c,+}(0) = \varepsilon_{c,-}(0) = 0$, and the Schur polynomial factorizes exactly as

$$\mathcal{P}(z, 0) = \left(z - \tilde\epsilon_\xi - \frac{r^2 W^2}{z} \right)^2 - r^4 \left\{ [\Delta^{(0)}_0]^2 - [\Gamma^{(0)}_{\rm PT}(\beta_0)]^2 \right\}. \tag{A3}$$

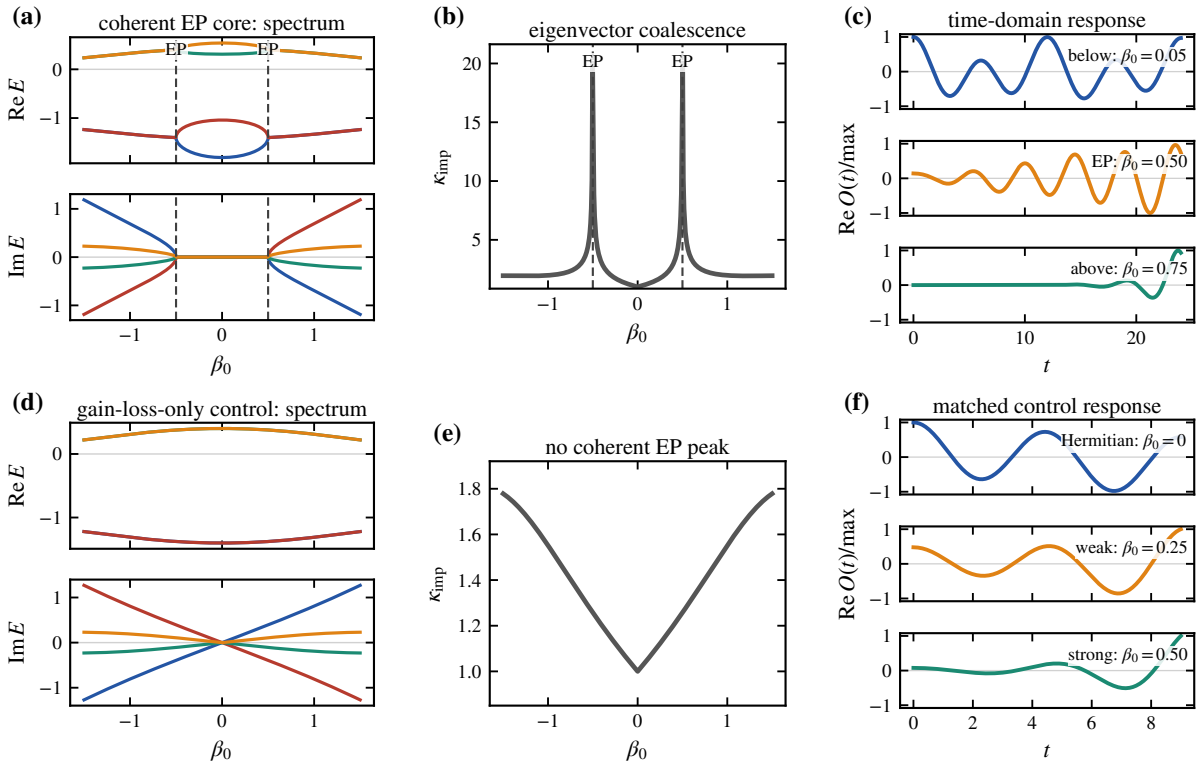


FIG. S1. **Frozen 4×4 compensated-core EP and matched control.** All panels use the rotated kernel of Eq. (22) at $k=0$ with $r=1$, $\tilde{\epsilon}_\xi=-1$, fixed scalar hybridization $W=0.75$, and $G=\Gamma_{\rm PT}=|\beta_0|$. The level $\tilde{\epsilon}_\xi=-1$ is an assigned benchmark coefficient; the historical label $\epsilon_\xi=-U/2$ at $U=2$ is not an interaction calculation. No microscopic bath DOS, temperature or self-consistent filling enters. Panels (a)–(c) use the coherent splitting $D=\Delta_{\rm eff}=0.50$. (a) Real and imaginary parts of all four eigenvalues. The exact local-core EPs are at $\beta_0=\pm0.50$. (b) The Euclidean right-eigenvector condition number. The plotted spacing $\Delta\beta_0=0.0075$ places its nearest positive sample at 0.5025, where $\kappa_{\rm imp}=20.0304$; the exact origin remains $\kappa_{\rm imp}(0)=1$. (c) Real part of the survival amplitude of main Sec. VIII below, at, and above the EP. Each trace is divided by its own maximum of $|\mathrm{Re}\,O|$ on the window: 1.00, 7.06, and 1.79×10^4 for $\beta_0=0.05$, 0.50, and 0.75. The normalized kernel contains the relative gain–loss but not the common width, so the amplitude grows linearly at the EP and exponentially above it; this growth belongs to the compensated representation and is not amplification by the passive physical kernel. Panels (d)–(f) repeat the same tests after switching off the coherent splitting ($D=0$); the divisors in (f) are 1.00, 2.09, and 12.9. The smooth condition number in (e) and the traces in (f) form the relative-gain–loss-only control; they are not interpreted as a coherent impurity EP or as a Kondo-scale enhancement.

The impurity-related $k=0$ double-root condition is therefore exactly the local core condition for arbitrary W and $r>0$.

Appendix B: Angular-Harmonic Motivation and the Defined Radial Kernel

We record the scalar angular selection rule motivating the auxiliary sector and then eliminate the auxiliary levels of the defined radial model. The fixed-axis SOC bath is not derived here from a time-reversal-symmetric two-dimensional Rashba host. A host-specific reduction would also require the spin-orbital hybridization vertices. The scalar selection rule alone establishes neither that reduction nor multiparticle integrability.

1. Angular Fourier Decomposition

In 2D polar momentum coordinates (k, θ), the bath fermion is decomposed into angular-momentum channels:

$$c_{k,\theta,\sigma} = \frac{1}{\sqrt{2\pi}} \sum_{m=-\infty}^{\infty} c_{k,m,\sigma}\, e^{im\theta}. \tag{B1}$$

The impurity at position $\mathbf{r} = 0$ (the lattice site) is local, so only the isotropic ($m = 0$) channel hybridizes with it:

$$c_{\mathbf{r}=0,\sigma} = \sum_k \frac{1}{\sqrt{2\pi}}\, c_{k,0,\sigma}. \tag{B2}$$

Higher harmonics $m \neq 0$ are decoupled from the impurity in the absence of anisotropy.

2. C_3 Selection Rule from $\cos 3\theta$

The trigonal anisotropy $\beta_3(k)k^3 \cos 3\theta$ is expanded as $\cos 3\theta = \frac{1}{2}(e^{3i\theta} + e^{-3i\theta})$. Substituting Eq. (B1) and performing the θ integral: $\int_0^{2\pi} \frac{d\theta}{2\pi}\, e^{i(m'-m)\theta}\, e^{\pm 3i\theta} = \delta_{m',m\mp 3}$. The trigonal term therefore couples only channels differing by $\Delta m = \pm 3$: $V_{\text{trig}} \propto \sum_{k,\sigma,m} c^\dagger_{k,m,\sigma} c_{k,m-3,\sigma} + \text{h.c.}$ The auxiliary fermions $O_{k,m,\sigma}$ represent precisely the $m = \pm 3, \pm 6, \ldots$ channels activated by this selection rule.

3. Gaussian Integration and Passive Relative Kernel

Since $H_O + H_{\text{hyb}}$ is bilinear in $O_{k,m,\sigma}$, integrating out all $m \neq 0$ modes is a Gaussian path integral within the quadratic auxiliary sector. For fixed (k, θ, σ) the auxiliary action has exactly the form

$$\begin{aligned} S_O + S_{cO} = &\int_{\mathcal{C}} dt dt'\, \bar{O}(t) A_O(t,t') O(t') \\ &+ \int_{\mathcal{C}} dt\, \left[\bar{c}(t) V_\sigma(t) O(t) + \bar{O}(t) V^\dagger_\sigma(t) c(t)\right]. \end{aligned} \tag{B3}$$

The Grassmann integral gives the exact quadratic influence action

$$S_{\text{infl}} = -\int_{\mathcal{C}} dt dt'\, \bar{c}(t) V_\sigma(t) g_O(t,t') V^\dagger_\sigma(t') c(t'), \tag{B4}$$

so the explicit retarded vertex contraction is

$$\Sigma^{O,R}_\sigma(t,t') = \sum_{m,m'\neq 0} V_{m\sigma}(t)\, g^{O,R}_{mm'}(t,t')\, V^*_{m'\sigma}(t'). \tag{B5}$$

No anomalous auxiliary propagator is used. For a number-conserving Hermitian auxiliary Hamiltonian this contraction contains the phase difference fixed by the vertex and its Hermitian conjugate. The Wigner step connecting this two-time expression to the frozen bath self-energy can be displayed explicitly. Set $T = t_{\text{av}} = (t+t')/2$ and $\tau = t - t'$, and write $\mathcal{V}_{mm'} = V_m V^*_{m'}$ for the $m = 0$-projected vertices of main Eq. (5). To first adiabatic order,

$$\begin{aligned} &V_{m\sigma}(T + \tau/2) V^*_{m'\sigma}(T - \tau/2) \\ &\quad = \mathcal{V}_{mm'} e^{i\zeta_\sigma [\phi(T+\tau/2) - \phi(T-\tau/2)]} \\ &\quad = \mathcal{V}_{mm'} e^{i\zeta_\sigma \tau \dot{\phi}(T)} \left[1 + O(\tau^3 \partial^3_T \phi)\right]. \end{aligned} \tag{B6}$$

For a stationary auxiliary propagator, its Wigner transform is consequently

$$\begin{aligned} \Sigma^{O,R}_\sigma(\omega, T) &= \Sigma^{O,R}_0[\omega + \zeta_\sigma \dot{\phi}(T)] \\ &= \Sigma^{O,R}_0(\omega) + \zeta_\sigma \dot{\phi}(T) \partial_\omega \Sigma^{O,R}_0(\omega) \\ &\quad + O[(\Omega/\epsilon_O)^2]. \end{aligned} \tag{B7}$$

Here the remainder denotes the relative gradient correction in the off-shell window $|\omega-\epsilon_O^{(m)}| \sim \epsilon_O$. Defining $\Sigma_0^{\mathrm{bath,R}} = \Sigma_{\mathrm{bg}}^R + \Sigma_0^{O,R}$ and $\Delta_{\mathrm{aux}} = \dot{\phi}(T)\partial_\omega \,\mathrm{Re}\,\Sigma_0^{O,R}$, therefore gives $\Sigma_\sigma^{\mathrm{bath,R}} = \Sigma_0^{\mathrm{bath,R}} + \zeta_\sigma \Delta_{\mathrm{aux}} + O[(\Omega/\epsilon_O)^2]$, with $\Delta_{\mathrm{aux}} \in \mathbb{R}$. For diagonal remote modes,

$$\begin{aligned}
\Sigma_0^{O,R}(\omega) &= \sum_{m\neq 0} \frac{|V_m|^2}{\omega - \epsilon_O^{(m)} + i0^+}, \\
\Delta_{\mathrm{aux}} &= -\dot{\phi}(T) \sum_{m\neq 0} |V_m|^2 \mathcal{P} \frac{1}{[\omega - \epsilon_O^{(m)}]^2} \\
&= -\dot{\phi}(T) \sum_{m\neq 0} |V_m|^2 \partial_{\epsilon_O^{(m)}} \mathcal{P} \frac{1}{\omega - \epsilon_O^{(m)}}.
\end{aligned} \tag{B8}$$

Thus the off-shell auxiliary sector produces a real principal-value correction in the low-energy window. If an auxiliary pole enters that window, the $-i\pi\delta(\omega - \epsilon_O^{(m)})$ part must instead be retained as ordinary on-shell bath damping. The same frozen shift follows after the unitary rotation $O_{k,m,\sigma} \to e^{-i\zeta_\sigma \phi(t)} O_{k,m,\sigma}$, which removes the drive phase from the vertices and shifts the auxiliary levels by $-\zeta_\sigma \dot{\phi}(t)$. The passive non-Hermitian component appears only after this shifted channel is transmitted through the retarded Dirac propagator. The point is clearest from the quadratic (d, c, O) inverse Green-function kernel,

$$\mathcal{G}^{-1} = \begin{pmatrix} g_d^{-1} & -W^\dagger & 0 \\ -W & g_c^{-1} & -V_\sigma \\ 0 & -V_\sigma^\dagger & g_O^{-1} \end{pmatrix}. \tag{B9}$$

Here W is static and all explicit drive dependence resides in $V_\sigma(t)$. Integrating out O and then c gives the two exact Schur complements

$$G_{c,\sigma}^{-1} = g_c^{-1} - V_\sigma g_O V_\sigma^\dagger, \qquad \Sigma_{d,\sigma} = W^\dagger G_{c,\sigma} W. \tag{B10}$$

Equivalently, $\Sigma_{d,\sigma} = W^\dagger g_c W + W^\dagger g_c (V_\sigma g_O V_\sigma^\dagger) g_c W + \cdots$, which explicitly displays the drive-dependent virtual path $d \xrightarrow{W} c \xrightarrow{V_\sigma(t)} O \xrightarrow{V_\sigma^\dagger(t')} c \xrightarrow{W^\dagger} d$. The order of the two Gaussian integrations is immaterial.

The first-order correction, its rigid-shift limit, and its exact SOC matrix form are Eqs. (9)–(11) of the main article. Here ω_0 denotes the low-energy evaluation point of the frozen kernel; at the compensation energy where $\partial_\omega \Lambda$ changes sign (of order D_{uv}/e for the linear Dirac width with a sharp cutoff), the Kramers–Kronig detuning vanishes, $\Delta_{\mathrm{coh}}(\omega_0) = 0$, while $\partial_\omega \bar{\Gamma}$ remains finite. For completeness, take the idealized symmetric Dirac width $\bar{\Gamma}(\epsilon) = g|\epsilon|\Theta(D_{\mathrm{uv}} - |\epsilon|)$. Its principal-value Kramers–Kronig partner is

$$\begin{aligned}
\Lambda(\omega) &= \frac{1}{\pi}\mathcal{P}\int_{-D_{\mathrm{uv}}}^{D_{\mathrm{uv}}} d\epsilon \, \frac{\bar{\Gamma}(\epsilon)}{\omega - \epsilon} \\
&= -\frac{g\omega}{\pi} \ln\left|\frac{D_{\mathrm{uv}}^2 - \omega^2}{\omega^2}\right|.
\end{aligned} \tag{B11}$$

Hence, for $|\omega| \ll D_{\mathrm{uv}}$,

$$\begin{aligned}
\partial_\omega \Lambda &= -\frac{2g}{\pi}\left[\ln\frac{D_{\mathrm{uv}}}{|\omega|} - 1\right] + O(\omega^2/D_{\mathrm{uv}}^2), \\
\partial_\omega \bar{\Gamma} &= g \,\mathrm{sgn}\,\omega.
\end{aligned} \tag{B12}$$

This shows explicitly both the compensation scale $|\omega_0| \simeq D_{\mathrm{uv}}/e$ in the low-energy form and the logarithmic growth of $|\Delta_{\mathrm{coh}}/\Gamma_{\mathrm{PT}}|$ near the Dirac node. For the exact sharp-cutoff expression with $D_{\mathrm{uv}} = \pi/4$, numerical solution of $\partial_\omega \Lambda = 0$ gives $|\omega_c| = 0.24608$, rather than the low-energy estimate $D_{\mathrm{uv}}/e = 0.28893$. A smooth cutoff shifts this number but not the mechanism. The SOC-split bath supplies the helicity channels and channel-resolved widths

$$\bar{\Gamma}_\eta(\omega) = \pi \sum_k |W_{k\eta}|^2 \delta[\omega - \varepsilon_{c,\eta}(k)]. \tag{B13}$$

Their difference enters the spin-mixing channel as $-i(\bar{\Gamma}_+ - \bar{\Gamma}_-)\sigma_x/2$, which is $\mathcal{PT}$-odd under $P = \sigma_x$. Exact $\mathcal{PT}$ symmetry with $P = \sigma_x$ therefore holds only for the normalized core of main Eq. (14). In the exact radial kernel,

where the drive-generated relative decay vanishes (Sec. B 4), the leading-order passive dimer is $\Delta_{\rm coh}\sigma_z + i\gamma\sigma_x$ with $\gamma = (\bar{\Gamma}_- - \bar{\Gamma}_+)/2$; it is $\mathcal{PT}$-symmetric with $P = \sigma_z$, and the real part of Σ_x plays the role of the detuning. For the normalized core the finite detuning acts as a causal floor:

$$|s|^4 = \left(\Delta_{\rm eff}^2 + \Delta_{\rm coh}^2 - \Gamma_{\rm PT}^2\right)^2 + 4\Delta_{\rm coh}^2\Gamma_{\rm PT}^2. \tag{B14}$$

Minimizing over the real control $\Delta_{\rm eff}^2 \geq 0$ gives

$$|s|_{\rm min} = \begin{cases} \sqrt{2|\Delta_{\rm coh}\Gamma_{\rm PT}|}, & |\Delta_{\rm coh}| \leq |\Gamma_{\rm PT}|, \\ \sqrt{\Delta_{\rm coh}^2 + \Gamma_{\rm PT}^2}, & |\Delta_{\rm coh}| > |\Gamma_{\rm PT}|. \end{cases} \tag{B15}$$

The square-root expression quoted for the near-compensated regime therefore has the necessary domain $|\Delta_{\rm coh}| \leq |\Gamma_{\rm PT}|$; outside it the minimum occurs at $\Delta_{\rm eff} = 0$. This is the sense in which the strictly Hermitian driven model generates the relative non-Hermitian kernel used in the main text. Compensation is local in frequency: for the audited values $\bar{\Gamma} = 0.12$, $\Gamma_{\rm PT} = 0.10$, the full sharp-cutoff kernel one common half-width from ω_c gives $|s|_{\rm min}/\Gamma_{\rm PT} = 1.0028$ on one side and 0.9004 on the other. Thus the finite-band problem has an order-$\Gamma_{\rm PT}$ avoided- coalescence floor across a linewidth.

The reduction relies on the following conditions:

(i) A local scalar angular vertex selects $m = 0$ before additional spin-orbital structure is specified. Here the retained two-component channel and its fixed-axis SOC are model assumptions, not a completed Rashba-host projection (main Sec. II).

(ii) Off-shell auxiliary harmonics, $\epsilon_O^{(m)} \gg D_{\rm uv}$ and $|V|/\epsilon_O^{(m)} \ll 1$, so the auxiliary contribution is a controlled principal-value shift. Residual on-shell weight gives ordinary bath damping rather than the relative $\mathcal{PT}$ channel.

(iii) Slow drive on the auxiliary memory time, $\Omega \ll \epsilon_O^{(m)}$, so the correction is organized in powers of Ω/ϵ_O.

(iv) A retarded relative-decay channel: an energy-dependent scalar width with $\partial_\omega\bar{\Gamma} \neq 0$, as in the cone-patch estimate, or, in the exact radial model where that conversion vanishes, the static helicity-width imbalance of Eq. (B13).

(v) H_O bilinear: no many-body approximation is made in the auxiliary-mode integration.

These conditions do not guarantee reachability of the normalized core; the leading mapping gives the bound quoted in the main article, which is incompatible with the off-shell quasistatic hierarchy.

4. Stationary rotating drive and full-momentum nonlinear-pole EP

The preceding obstruction concerns the oscillatory quasistatic protocol. A stationary alternative follows from the same microscopic vertex by taking $\phi(t) = \Omega t$. After the auxiliary rotation of main Sec. II, the remote levels are shifted by the constant $-\zeta_\sigma\Omega$, so the rotated quadratic Hamiltonian is exactly time independent. For one remote Hermitian level with vertex V_O, its spin-even and spin-odd contributions to the bath inverse are

$$B_{\rm raw}(z) = \frac{V_O^2(z - \epsilon_O)}{(z - \epsilon_O)^2 - \Omega^2}, \qquad A(z) = -\frac{V_O^2\Omega}{(z - \epsilon_O)^2 - \Omega^2}. \tag{B16}$$

The constant spin-even principal-value shift is absorbed into the bath chemical potential, $B(z) = B_{\rm raw}(z) - B_{\rm raw}(0)$. At fixed controls this is an explicitly chosen scalar bath-energy offset. It does not alter the auxiliary spin-odd term. When V_O or Ω changes, maintaining this convention changes the compensating scalar offset as well; all certificates and scans use that recentered family of Hamiltonians.

For $\alpha_{\rm b} = 1$, scalar momentum-independent impurity hybridization W, and radial cutoff K, the physical-sheet impurity self-energy is

$$\Sigma_d^{\rm I}(z) = W^2 \int_0^K \frac{k\,dk}{2\pi} \left[(z - B - k^2)\mathbb{I} - \lambda k\sigma_x - A\sigma_z\right]^{-1}. \tag{B17}$$

The integral is evaluated analytically and independently checked by adaptive quadrature. Every open helicity channel must be continued. If k_j is an on-shell radial root on the real cut, define

$$P_j = \frac{1}{2}\left[\mathbb{I} + \frac{\lambda k_j \sigma_x + A\sigma_z}{z - B - k_j^2}\right],$$
$$\rho_j = \frac{k_j}{2\pi|v_j|}P_j, \qquad v_j = 2k_j + \frac{\lambda^2 k_j}{z - B - k_j^2}. \tag{B18}$$

Analytic continuation of the roots and projectors then gives the outgoing resonance sheet

$$\Sigma_d^{\mathrm{II}}(z) = \Sigma_d^{\mathrm{I}}(z) - 2\pi i W^2 \sum_{j\in\mathrm{open}} \rho_j(z). \tag{B19}$$

At the solution below there are two open channels. Continuing only one cut misses the EP.

With F defined in Eq. (16), the nonlinear EP conditions are $F(z_{\mathrm{EP}}) = \partial_z F(z_{\mathrm{EP}}) = 0$ and $\partial_z^2 F(z_{\mathrm{EP}}) \neq 0$, together with geometric multiplicity one of the inverse kernel. To certify the zero without evaluating a singular eigenbasis, for each point on a closed control loop in (ϵ_d, V_O) we branch-track the nearby critical point z_c satisfying $\partial_z F(z_c) = 0$ and form $\mathcal{D}_{\mathrm{loc}}(\epsilon_d, V_O) = F(z_c; \epsilon_d, V_O)$. A degree interpretation assumes that this critical-point branch extends smoothly over the enclosed domain with $\partial_z^2 F \neq 0$ there. The implicit-function theorem supplies a local neighborhood of a nondegenerate critical point; checking only the boundary is insufficient for a global extension. Under this domain assumption, and provided $\mathcal{D}_{\mathrm{loc}}$ is nonzero on the boundary, its integer $W_C = \frac{1}{2\pi}\Delta_C \arg \mathcal{D}_{\mathrm{loc}}$ is the Brouwer degree of the smooth map $(\epsilon_d, V_O) \mapsto (\mathrm{Re}\,\mathcal{D}_{\mathrm{loc}}, \mathrm{Im}\,\mathcal{D}_{\mathrm{loc}})$. A nonzero degree forces at least one enclosed zero; regular zeros contribute the sign of the corresponding real 2×2 Jacobian. At a zero satisfying $F = \partial_z F = 0$ and $\partial_z^2 F \neq 0$, the coalescence is quadratic, while the rank-one inverse-kernel test below excludes a scalar or diabolic degeneracy. Explicit tracking of the two poles supplies the equivalent EP2 monodromy test: one circuit exchanges their labels and their separation has half-integer winding.

For

$$\begin{aligned} \lambda = 0.05, \quad W = 0.10, \quad K = \pi/4, \\ \epsilon_O = 120, \quad \Omega = 60, \quad T_{\mathrm{b}} = 0.1, \end{aligned} \tag{B20}$$

the solution is

$$\begin{aligned} z_{\mathrm{EP}} &= 0.011972834346 - 0.002807216274i, \\ V_O &= 1.213422009614, \qquad \epsilon_d = 0.015170349783. \end{aligned} \tag{B21}$$

The algebraic residuals are $|F| = 1.44 \times 10^{-21}$, $|\partial_z F| = 1.09 \times 10^{-15}$ and $\partial_z^2 F = 1.686448434 - 0.190336728i$. The two singular values of the inverse kernel are 1.207×10^{-3} and 1.22×10^{-18}, establishing geometric multiplicity one. On the reference ellipse $\delta\epsilon_d = 10^{-4}$ and $\delta V_O = 10^{-2}$, $W[\mathcal{D}_{\mathrm{loc}}] = +1$; the two poles exchange and their separation has half-winding $+1/2$. The minimum boundary modulus is 2.10×10^{-9}, the largest phase step is $0.0882 < \pi/2$, and $|\partial_z^2 F|$ remains above 1.697. The maximum pole-tracking step is only 0.103 of the local half-separation, and the maximum pole residual is 8.62×10^{-21}. Halving and doubling both ellipse radii leave the winding and pole exchange unchanged.

The causal and microscopic controls are independent of the root solve. The analytic radial integral agrees with direct quadrature to 7.93×10^{-19}. Matching the upper rim of the physical sheet to the continued lower rim gives a two-cut residual 4.02×10^{-9} at $\eta = 10^{-8}$. At $\omega = \mathrm{Re}\, z_{\mathrm{EP}}$ the physical half-width matrix is $-\mathrm{Im}\,\Sigma_d^{\mathrm{I}} = 0.0025\,\mathbb{I} - 5.963 \times 10^{-4}\sigma_x + 0\,\sigma_z$. Its eigenvalues are positive, 0.00190368 and 0.00309632 (half-widths; the rate eigenvalues of $i(\Sigma^R - \Sigma^A)$ are twice these), so $\bar{\Gamma} = 0.00250000$ and $\Gamma_{\min}/\bar{\Gamma} = 0.76147$. The driven positive $k = 0$ threshold is $+0.00817959$; the certified pole is $0.00379325 = 1.351|\,\mathrm{Im}\, z_{\mathrm{EP}}|$ above it. Thus both open cuts, the threshold proximity, and the lower-branch van Hove structure are already included in the continuation. Finally,

$$\begin{aligned} \frac{V_O}{\epsilon_O - \Omega} = 0.02022, \qquad \frac{\epsilon_O - \Omega}{T_{\mathrm{b}}} = 600, \qquad \frac{T_{\mathrm{b}}}{\bar{\Gamma}} = 40, \\ \frac{\Omega}{\bar{\Gamma}} = 2.40 \times 10^4, \quad \frac{\Omega}{\alpha_{\mathrm{b}} K^2} = 97.27. \end{aligned} \tag{B22}$$

Thus the auxiliary sector is off shell and thermally empty, while its exact frequency dependence has been retained rather than expanded in Ω/ϵ_O. The large $\Omega/\bar{\Gamma}$ ratio would invalidate an oscillatory quasistatic interpretation but is irrelevant to the exactly stationary rotating-frame Hamiltonian. The certificate itself is retarded and temperature

independent; linewidth-resolved equilibrium observables for this witness require $T_{\rm b} \lesssim 10^{-3}$. Figure 2 displays the local winding and pole exchange.

The same equations can be continued in SOC with W held fixed at 0.10 while retuning the real impurity gate and auxiliary vertex. We obtain a regular one-dimensional locus for $0.02 \le \lambda \le 0.1578$. Representative points are

λ	$\mathrm{Re}\, z_{\rm EP}$	$\mathrm{Im}\, z_{\rm EP}$	ϵ_d	V_O
0.05	0.01197283	−0.00280722	0.01517035	1.21342201
0.10	0.02035701	−0.00292855	0.02323413	1.83318448
0.15	0.03265022	−0.00303645	0.03515182	2.41589290

At all three points $\mathcal{D}_{\rm loc}$ has winding +1, the endpoint pole permutation is (12), and the pole-separation half-winding is +1/2. The maximum tracking-step/half-separation ratios are 0.103, 0.100, and 0.119, with no failed pole solves. The corresponding threshold gaps in pole widths are 1.351, 0.577, and 0.076 and the passivity margins are 0.761, 0.614, and 0.515, respectively. Throughout the continued segment $V_O/(\epsilon_O - \Omega) < 0.042$ and the physical rate matrix remains positive. As shown in main Fig. 3, the branch approaches the upper open-channel threshold near the high-SOC end while its passivity margin remains positive. This threshold proximity is distinct from a passivity failure in a separate normalized sharp-cutoff audit. We do not classify its endpoint and do not infer a two-dimensional EP phase region.

This certificate concerns the full curved retarded kernel only; it is not identified with the nilpotent constant-matrix EP, and no full-momentum Yang–Baxter structure is inferred from it. The same SOC off-diagonal dressing supplies the coherent mixing channel through $\Delta_{\rm eff} = \sum_k |W_k|^2 \,\mathrm{Re}\, g_x$, as defined in main Sec. III B. It does not convert the drive into a relative decay. With both helicity roots open, write $r_\pm = (\lambda^2 k_\pm^2 + A^2)^{1/2}$ for the two on-shell roots of Eq. (B18). The on-shell condition gives $r_- - r_+ = \lambda^2$, so both roots share $\mathcal{D} = 2r_+ + \lambda^2 = 2r_- - \lambda^2$ and have radial densities $k_\pm/|v_\pm| = r_\pm/\mathcal{D}$. The σ_z weight $\pm A/(2r_\pm)$ of each projector is therefore offset by its density: $\mathrm{Im}\,\Sigma_z = 0$ on the physical sheet to all orders in A, and the identity part gives the flat width $\bar\Gamma = W^2/4$. At first order this is the statement $\mathrm{Im}\sum_k |W_k|^2 a_k b_k = \mathrm{Im}\sum_k |W_k|^2 g_{x,k}^2 = 0$ in main Eq. (11). At $\omega = \mathrm{Re}\, z_{\rm EP}$ the drive enters as the real coherent coupling $\mathrm{Re}\,\Sigma_z = 6.496 \times 10^{-4}$, and the relative decay is the σ_x part 5.963×10^{-4} of the half-width matrix above. With the drive off it is $5.568 \times 10^{-4} = W^2\lambda/(4\alpha_{\rm b}\Delta) = \pi W^2(\rho_- - \rho_+)/2$, the static SOC helicity imbalance. The EP needs both: for $V_O = 0$ the coherent coupling is absent. A drive-generated σ_z loss appears only below the driven threshold $\omega = 0.0082$, where one helicity band is open ($\mathrm{Im}\,\Sigma_z = -1.24 \times 10^{-3}$ at $\omega = 0$), or with a momentum-dependent vertex.

a. Result for the defined radial model. Gaussian elimination leaves the chosen radial channel, labelled by k, with a frequency-dependent matrix influence kernel. This is exact for the specified model; it does not complete a host-specific angular reduction. The exact positive-energy radial densities are given in the main article [Eq. (2)]; for $-\lambda^2/(4\alpha_{\rm b}) < \omega < 0$ the lower branch instead gives $\rho_-(\omega) = \lambda/[2\pi\alpha_{\rm b}(\lambda^2 + 4\alpha_{\rm b}\omega)^{1/2}]$, with a radial van Hove edge at $-\lambda^2/(4\alpha_{\rm b})$.

The radial bath has the exact DOS above and couples to the impurity through the channel-preserving vertex of Eq. (B2). The Hadamard rotation diagonalizes the bare SOC bath and impurity–bath vertex, but it does not diagonalize the drive-dressed scattering kernel. In the separate constant-core benchmark of main Sec. IV B, this distinction appears as the $i\Gamma_{\rm PT}$ matrix element converting between the two chiral channels. Section IX therefore first uses eigenphases of the full 2×2 unitary scattering matrix only for exact one-particle phase counting. Multiparticle factorization is asserted only after the additional equal-velocity branchwise linearization and folding stated explicitly there.

Appendix C: Finite-U Matrix Block-Wilson Continuation

This section tests the interaction stability of the nonlinear causal pole EP of Sec. B 4. It is separate from the scalar-hybridization Markov-core residue audit in Sec. L. The present calculation retains the SOC-resolved 2×2 physical-bath hybridization, the analytic two-cut continuation, and the exact frequency dependence of the remote auxiliary level. The low-energy physical bath is represented by a finite matrix Wilson chain; the remote level remains in the analytic one-body kernel.

1. Matrix chain and exact finite-chain response

For the $\lambda = 0.05$, $W = 0.10$ witness of Eq. (B20), the positive matrix spectral measure is logarithmically discretized with Wilson parameter $\Lambda = 1.5$ and mapped by block Lanczos to a chain with 2×2 onsite and hopping blocks. We use

12 chain sites for the endpoint center and pointwise certificate, then repeat the complete-basis calculation at $N = 10$ and 12 for a common-contour chain-length test. Each length averages four independent logarithmic discretization shifts. Star-to-chain reconstruction errors lie between 3.5×10^{-15} and 2.3×10^{-14}.

The impurity interaction is $H_U = U n_{d\uparrow} n_{d\downarrow}$. Each finite chain is solved in untruncated fixed-charge sectors for its ground state and $N \pm 1$ correction vectors. Writing $q_\alpha = n_{d\bar{\alpha}} d_\alpha$, we use

$$G_{\alpha\beta}(z) = \langle\langle d_\alpha; d_\beta^\dagger \rangle\rangle_z,$$
$$F_{\alpha\beta}(z) = \langle\langle q_\alpha; d_\beta^\dagger \rangle\rangle_z. \tag{C1}$$

The equation-of-motion identity is exact for each untruncated finite chain. For the reported mesh-averaged embedding, F and G are first averaged over the four chains and their ratio defines the estimator

$$\Sigma_U(z) = U F(z) G^{-1}(z). \tag{C2}$$

An independent Dyson construction agrees with Eq. (C2) to maximum absolute residual 4.67×10^{-16} and maximum relative residual 1.32×10^{-11}. At $U = 0$, the exact chain response agrees with the direct one-body Green function to 2.02×10^{-14}; anticommutator sum rules are satisfied at machine precision.

The determinant F_U is obtained by adding Eq. (C2) to the same analytic outgoing-sheet one-body kernel, Eq. (B19). The complex resonances belong to this embedding, not to the spectrum of the closed Hermitian chain. Exactness of the finite-chain correlators does not make this an exact interacting continuum self-energy. Simultaneously solving $F_U = \partial_z F_U = 0$ while continuing from $U = 0$ gives

U	$\mathrm{Re}\, z_{\mathrm{EP}}$	$\mathrm{Im}\, z_{\mathrm{EP}}$	ϵ_d	W
0	0.01197283	-0.00280722	0.01517035	0.10000000
0.0010	0.01189782	-0.00255800	0.01483973	0.09577738
0.0015	0.01185780	-0.00244927	0.01468704	0.09385894
0.0020	0.01181724	-0.00234964	0.01454259	0.09205708
0.0025	0.01177670	-0.00225815	0.01440607	0.09036305

The $U = 0.0025$ endpoint is reached both by direct initialization and by stepping from $U = 0.0020$. All four discretization shifts retain ground charge two. The interaction is 4.05×10^{-3} of the bandwidth $\alpha_{\mathrm{b}} K^2 = 0.617$; with the retuned local width $\bar{\Gamma}_{\mathrm{self}} = W^2/4 = 0.0020414$, $U/\bar{\Gamma}_{\mathrm{self}} = 1.225$. The continued level $\epsilon_d = 0.0144$ lies about $7\bar{\Gamma}_{\mathrm{self}}$ above the chemical potential, so the impurity is nearly empty. The endpoint is therefore a weakly correlated resonance with U comparable to the local width, far from a local-moment, bandwidth-scale, or Kondo strong-coupling limit.

a. Direct filling check. We evaluate occupations in the $T = 0$ grand-canonical ground ensemble of the Hermitian low-band reference at $\mu = 0$, with the same four archived meshes $z_W = 0.125, 0.375, 0.625, 0.875$. The symbol z_W labels logarithmic mesh shifts; z elsewhere is a complex frequency. The arithmetic average of occupations is taken after solving each chain. Define $D_d = \langle n_{d\uparrow} n_{d\downarrow} \rangle$ and $P_1 = \langle n_d \rangle - 2D_d$.

N	U	$\langle n_d \rangle$	D_d	P_1
10	0	0.021514	0	0.021514
10	0.0025	0.017924	0	0.017924
12	0	0.020652	9.65×10^{-7}	0.020650
12	0.0025	0.017064	5.73×10^{-7}	0.017063

The $U = 0$ rows use Eq. (B21); both interacting rows use the reported $N = 12$ endpoint controls, to compare filling at fixed controls rather than at separately retuned centers. At $N = 12$, $U = 0.0025$, the four occupancies range from 0.015847 to 0.018766; the mean empty-state probability is 0.982937. The vanishing $N = 10$ double occupancy reflects its ground-charge sector, not a developed local moment. These reference-state measurements establish that the reported endpoint is nearly empty. They neither determine the laboratory driven state nor establish thermodynamic filling convergence. Separate occupied block-Wilson scans can locate double zeros in individual finite-chain embeddings, but their centers and fillings are bath dependent; they have not established a converged interacting continuum EP. The diagnostics below are not used to upgrade the low-filling interaction claim.

b. Continuum comparison. The same $T = 0$, $\mu = 0$ filling evaluated on the continuum bath by the imaginary-axis integral of App. N is $\langle n_d \rangle = 0.03468$ at the $U = 0$ controls of Eq. (B21), against 0.020652 for the $N = 12$ chains. At the respective $U = 0.0025$ EPs of the (ϵ_d, W) continuation the continuum value is 0.02407. The one-body $U = 0$ ground states of the $N = 10$ and 12 chains contain one and two electrons. The chains therefore under-resolve the occupied bath, which the $N = 10 \to 12$ comparison of App. C does not detect. App. N 6 shows a substantial bath dependence of the KR interaction shift. The ED/KR comparison also changes the mesh estimator, so it does not isolate the interaction-approximation error.

2. Occupied-bath convergence diagnostics

Additional tests retain the same stationary bath and separate pole-finding accuracy from bath convergence. At $U = 0.0005$ and $\mu = 0.012$, fixing $V_O = 1.21342201$ and retuning (ϵ_d, W) gives the following finite-embedding centers. Here N_b is the number of two-component bath blocks; these meshes differ from the four historical low-band meshes above.

N_b	ϵ_d	W	$\langle n_d \rangle$
6	0.01620455	0.12234042	1.418855
8	0.01819760	0.13144545	0.774442
10	0.01329704	0.10121308	1.238022

All three centers pass their pointwise double-zero checks. The six- and eight-block cases also pass two-sided pole reconstruction and refined exchange loops. The variation of the controls and fillings precludes a continuum conclusion. Averaging two shifts and refining the radial mesh at eight blocks does not remove the length dependence.

A complementary test optimizes passive finite baths with positive matrix residues and constraints on occupied spectral moments. At $U = 0.2$, $\mu = 0.012$, $W = 0.10$ and $\epsilon_d = 0.01517035 - U/2$, baths with 8, 10 and 12 individual fermion orbitals give, respectively, $\langle n_d \rangle = 0.993998, 0.994036, 0.994052$ and $D_d = 0.00131355, 0.00132975, 0.00132698$. These are fixed-control reference states, not retuned EPs; the gate prescription is not an exact particle–hole symmetry of the asymmetric bath. Despite the static stability, the maximum ten-to-twelve-orbital self-energy change on five upper-half-plane probes is about $0.33\bar{\Gamma}_0$, and changing fit weights produces larger differences. Spin-density orientation and the associated Hartree matrix remain sensitive to the bath representation. The fitted baths also retain one-body hybridization errors. These tests show why charge suppression alone is insufficient for a pole conclusion.

Physical-sheet causality and coalescence are independent checks: selected split-pole references pass sampled positive-rate and spectral-matrix tests. These finite grids do not prove positivity everywhere. Finally, evaluation of a finite-chain rational self-energy in the lower half-plane defines the reported embedding; convergence on the physical sheet does not by itself establish the outgoing continuation of the interacting continuum self-energy. That continuation and bath convergence remain open tasks. The full inputs, fitting objectives and diagnostics accompany the revision.

3. Pointwise topological gate

At the interacting center we evaluate a 64-point ellipse with half-widths $\delta\epsilon_d = 10^{-4}$ and $\delta W = 10^{-3}$. At every point the nearby critical point satisfying $\partial_z F_U = 0$ and the two associated poles are independently branch tracked. The resulting gates are

Diagnostic	Result
Local-discriminant winding	-1.00000000000012
Endpoint pole permutation	(12)
Pole-separation half-winding	-0.50000000000005
Minimum boundary modulus	1.5060×10^{-9}
Maximum phase increment	0.27725
Minimum $\lvert\partial_z^2 F_U\rvert$	1.71153
Maximum $\lvert\partial_z F_U\rvert$	1.1150×10^{-15}
Minimum pole separation	8.3767×10^{-5}
Maximum pole residual	2.5452×10^{-21}
Assignment/half-separation ratio	$0.42415 < 1/2$
Maximum anticommutator sum-rule error	3.7682×10^{-15}

Thus the contour is nondegenerate, its phase sampling is resolved, the critical-point branch remains quadratic, and the pole assignment satisfies the explicit separation guard. The resolved winding and pole exchange in Fig. 5 support the independently located interacting double zero. A degree -1 follows under the domain assumption stated in App. B 4; no unsigned zero count or interval-arithmetic enclosure is asserted.

As a discretization-center check, increasing the average from four to eight shifts moves $\mathrm{Re}\, z_{\mathrm{EP}}$, $\mathrm{Im}\, z_{\mathrm{EP}}$, ϵ_d, and W by 0.022%, 0.71%, 0.14%, and 0.33%, respectively. This is a center-convergence check, not an infinite-chain extrapolation.

4. Common-contour chain-length gate

The $N = 10$ and $N = 12$ interacting centers are solved independently rather than inherited from one another. Their maximum absolute coordinate shift is 8.737×10^{-5} and their maximum relative coordinate shift is 3.746×10^{-3}. We then evaluate both chains on the same ellipse, centered on the reoptimized $N = 12$ point, with half-widths $\delta\epsilon_d = 2 \times 10^{-4}$ and $\delta W = 2 \times 10^{-3}$. At 256 contour points the branch-tracked local discriminant D_N gives the complete-basis gates

Diagnostic	$N = 10$	$N = 12$
Winding	-1	-1
Half-winding	$-1/2$	$-1/2$
Pole permutation	(12)	(12)
Sampled $\min_j \lvert D_{N,j}\rvert$	2.417×10^{-9}	2.897×10^{-9}
Assignment ratio	0.1388	0.1352
Sum-rule error	1.256×10^{-15}	6.280×10^{-15}

Here "assignment" is the maximum branch-assignment step divided by the local half-separation. It remains well below 1/2 for both chains. The sampled comparison is

$$\begin{aligned} \max_{j=1,\ldots,256} |D_{12,j} - D_{10,j}| &= 1.638 \times 10^{-9} \\ &< \min_{j=1,\ldots,256} |D_{10,j}| = 2.417 \times 10^{-9}, \end{aligned} \tag{C3}$$

giving ratio $0.6775 < 1$. If this strict inequality held uniformly on the continuous boundary, the straight-line homotopy would avoid zero and, assuming smooth extensions over the interior, the two maps would have the same Brouwer degree. Here the extrema are sampled, without a rigorous interpolation-error bound. The 128-point calculation gives the same winding, exchange and half-winding, with ratio 0.6758; doubling the resolution halves the assignment ratios and changes this ratio by 0.25%. These are resolved numerical stability checks for the specified embeddings. The holomorphic zero-count version of Rouché's theorem does not apply to these maps of two real controls. Nor does the comparison bound the omitted Wilson-chain tail or prove thermodynamic convergence.

5. Kept-space calibration and scope

To quantify what an ordinary truncated run would resolve, we compare kept-space responses with the exact response on the identical four 12-site chains:

N_{keep}	max. G	max. F	max. Σ_U	sum-rule deficit
320	11.67%	47.35%	38.00%	2.61%
640	11.76%	32.90%	24.74%	1.51%
960	3.71%	4.57%	4.18%	1.40%

The retained deviations decrease at $N_{\mathrm{keep}} = 960$, but this comparison is not used to certify the topology: the load-bearing result is the untruncated complete-basis response on the specified finite chains.

Chain-length and Λ convergence to the continuum still require a complete-basis spectral NRG/DM-NRG implementation; App. N compares bath dependence within a finite-U saddle evaluated on these chains and on the continuum; it does not bound the exact continuum error. A short-chain four-band discretization including the remote auxiliary pair selects ground charges $1, 2, 2, 1$ across the four shifts; its center is contaminated by finite-size charge steps and is reported neither as a pass nor as a failure.

Appendix D: Keldysh Mean-Field Theory

The slave-boson representation $d_\sigma^\dagger = f_\sigma^\dagger b$ introduces auxiliary spinon fermions f_σ and a slave boson b (enforcing no double occupancy) with condensate $b_c = \langle b \rangle$ and Lagrange multiplier δ_c enforcing the constraint $r^2 + \sum_\sigma f_\sigma^\dagger f_\sigma = Q$ (where $r = |b_c|$). Here $Q = 1$ is the physical spin-1/2 constraint, while $Q = N q_{\mathrm{occ}}$ denotes the large-N or channel-rescaled normalization used in the corresponding numerical saddle. The quadratic spinon action can be organized

either with an explicit bath or with its complete influence kernel. In the latter representation the retarded impurity kernel has the schematic form

$$\mathcal{G}_{ff}^{R,-1} = \omega - \epsilon_d - \delta_c - r^2 D_0(\beta_0)\sigma_x - r^2 K_\Gamma^R(\omega, \beta_0), \tag{D1}$$

and its lesser partner contains $r^2 K_\Gamma^<$. Therefore $\partial_r(rW) = W$, $\partial_r(r^2 D_0) = 2rD_0$ and $\partial_r(r^2 K_\Gamma) = 2rK_\Gamma$. The first derivative is used when the bath is explicit. The latter two are used in the integrated-bath representation employed by the numerical Markov saddle; the two descriptions are alternatives for a given bath. For the $U \to \infty$ Coleman saddle, variation of the complete quadratic contour action gives the basis-independent equations

$$0 = r^2 + n_f - Q, \qquad n_f = -i \int \frac{d\omega}{2\pi} \mathrm{Tr}_s G_{ff}^<(\omega), \tag{D2}$$

$$0 = 2r\delta_c + 2rD_0\,\mathcal{X} + 2r\mathcal{I}_\Gamma. \tag{D3}$$

For comparison the alternative explicit-bath force is $\mathcal{F}_W$; the integrated equation above uses only $\mathcal{X}$ and $\mathcal{I}_\Gamma$:

$$\mathcal{F}_W = -i \int \frac{d\omega}{2\pi} \sum_k \mathrm{Tr}_s \left[V_H G_{cf,k}^< + V_H^\dagger G_{fc,k}^< \right], \tag{D4}$$

$$\mathcal{X} = -i \int \frac{d\omega}{2\pi} \mathrm{Tr}_s [\sigma_x G_{ff}^<], \tag{D5}$$

$$\mathcal{I}_\Gamma = \mathrm{Re}\left\{ -i \int \frac{d\omega}{2\pi} \mathrm{Tr}_s \left[K_\Gamma^R G_{ff}^< + K_\Gamma^< G_{ff}^A \right] \right\}. \tag{D6}$$

The last line is the Langreth projection of the derivative of the dissipative influence action. Dropping $K_\Gamma^< G^A$ generally produces a complex force and is not a valid thermal saddle. If the relevant bath is retained explicitly instead, $\mathcal{I}_\Gamma$ is absent as a separate term because it is already contained in $\mathcal{F}_W$; the two descriptions must not be combined. Equation (D3) is divided by $2r$ only after imposing a finite-r guard. At the $r = 0$ boundary the undivided equation is used, so the numerical update never introduces a spurious $1/r$ singularity. Under the Hadamard rotation, $\sigma_x \mapsto \sigma_z$, $\sigma_z \mapsto \sigma_x$, and $V_H \mapsto W_k I$. Hence in the chiral basis

$$\mathcal{F}_W^\pm = -i \int \frac{d\omega}{2\pi} \sum_{k,\eta} W_k \left(G_{c_\eta f_\eta, k}^< + G_{f_\eta c_\eta, k}^< \right), \tag{D7}$$

$$\mathcal{X}_\pm = -i \int \frac{d\omega}{2\pi} (G_{++}^< - G_{--}^<), \tag{D8}$$

while the relative dissipative force depends on the off-diagonal chiral components because $\widetilde{K}_\Gamma \propto iG\sigma_x$. The full matrix lesser Green function must therefore be retained. The lesser Green's functions are evaluated from the bath sandwich $G^< = G^R \Sigma_{\mathrm{bath}}^< G^A$, with $\Sigma_{\mathrm{bath}}^< = if\Gamma_{\mathrm{bath}}$. When the complete thermal influence kernel is retained this is algebraically equivalent to $G^< = f(\omega)[G^A - G^R]$ [1]; the sandwich form is used in the solver so that the channel matrix and charge-continuity test remain explicit. This is an instantaneous-equilibrium closure at leading adiabatic order. Corrections of order $O[(\Omega/\epsilon_O)^2]$ and the corresponding Wigner gradients would encode drive-induced redistribution; no energy-pumping result is asserted at the retained order. For the stationary rotating-drive witness the Hamiltonian transformation is exact, but it does not establish a thermal density matrix. A common Fermi function requires an equilibrating reservoir or a suitable rotating-frame equilibrium preparation. Negligible occupation of the remote levels removes one source of lesser weight; it does not by itself prove thermalization of the retained sector. The retarded $U = 0$ certificate does not require this thermal closure. The constraint multiplier δ_c is real in the gauge-fixed saddle, as required by its Hermitian constraint field; this restriction is not equivalent to imposing retarded boundary conditions.

The finite-$U = 2$ Schrieffer–Wolff expressions used later are a separate frozen estimate and are not inserted into this saddle.

1. Low-temperature self-consistent gate

The corrected $U \to \infty$ saddle was solved for the fixed-W small-width path with the complete thermal Markov influence kernel. At $T_{\mathrm{b}} = 10^{-6}$ a finite-r solution exists over $0.35 \le \beta_0 \le 0.55$. The amplitude decreases smoothly from $r = 0.008754$ to 0.007956; at the core point $\beta_0 = 0.50$, $r = 0.008203$ and $\delta_c - 1 = 8.75 \times 10^{-10}$. These values replace those of the archived solver, whose uniform base frequency grid, refined only near the poles, under-resolved

the logarithmic occupied-band integral and overestimated r by 19%; its pole-centered refinement test could not detect this. At $T_{\rm b}=0$ the finite-cutoff integrals have logarithmic primitives. In the deep-level, wide-band limit $\tilde{\epsilon}_\xi \to 0$, the spinon poles are $z_\pm = -ir^2\Gamma_0 \pm r^2 s_0$ with $s_0^2 = D^2 - G^2$, every saddle integral reduces to $L(z) = \ln(-z) - \ln(-D_{\rm b} - z)$ at $z_\pm$, where $D_{\rm b} = 4$ is the band half-width of this path, and the corresponding wide-band mean-field asymptotic is

$$r^2\Gamma_0\sqrt{1+u} = T_K^{(0)} e^{F(u)}, \qquad T_K^{(0)} = D_{\rm b}\, e^{-\pi|\epsilon_\xi|/(2\Gamma_0)}, \qquad u = s_0^2/\Gamma_0^2, \tag{D9}$$

with $F(u) = \sqrt{u}\arctan\sqrt{u}$ for $u \geq 0$ and $F(u) = -\sqrt{-u}\,\mathrm{artanh}\sqrt{-u}$ for $-1 < u < 0$. The real expression is restricted to $u > -1$ and the passive kernel domain. It is a mean-field width scale, not an exact physical Kondo temperature. It reproduces the numerical $T_{\rm b} = 0$ saddle to the digits shown ($r = 0.008298$ at the core point). An independent finite-cutoff calculation gives $r = 0.00820293449$ at $T_{\rm b} = 10^{-6}$ and the EP. Three separately solved grids refine both the logarithmic tail and the uniform base mesh. At $\beta_0 = 0.35, 0.50, 0.55$, the finest-grid relative error in r is below 8.9×10^{-7} and the independently evaluated saddle residual below 1.4×10^{-7}. These checks concern the scalar constant-core saddle, distinct from the continuum matrix KR equations of App. N. Since $F(u) = u - u^2/3 + O(u^3)$ is analytic, the slave-boson scale $\sqrt{|z_+ z_-|} = r^2\Gamma_0\sqrt{1+u} = T_K^{(0)} e^{F(u)}$ has neither a cusp nor an enhancement at the EP, where it equals the core-free value $T_K^{(0)} = 8.26 \times 10^{-6}$.

As shown in App. A, the core EP stays exactly at $\beta_0 = 0.50$ for every finite r, and no interior finite-k EP is found.

At the reference temperature $T_{\rm b} = 0.1$, the same closure has no finite-r solution on this parameter path and reaches the $r = 0$ boundary. This is why the $T_{\rm b} = 0.1$ spectra remain frozen diagnostics and why no finite-temperature condensate or Kondo-scale enhancement is inferred. The saddle values above are obtained from the complete contour equations (D2)–(D3); they do not use the $T_{\rm b} = 0$ estimate of App. G.

Appendix E: Bath-Controlled FDR and Physical Spectral Function

The complete reservoir-coupled problem uses the Keldysh lesser Green's function fixed by the thermal bath. No independent mode-by-mode occupation prescription enters the physical occupations or saddle-point equations.

a. Physical bath construction. With $G^<_{ij}(t,t') = i\langle d_j^\dagger(t') d_i(t)\rangle$ and hence $n_f = -i\,\mathrm{Tr}\, G^<(t,t)$,

$$\begin{aligned} \Sigma^<_{\rm bath} &= i f \Gamma_{\rm bath}, \\ G^<_{\rm bath} &= G^R \Sigma^<_{\rm bath} G^A = f(G^A - G^R) \end{aligned} \tag{E1}$$

follows directly from the Keldysh identity for a system bilinearly coupled to a thermal reservoir [1]. For the uniform reference $\epsilon_\xi = -U/2 = -1$, $U = 2$, and $T_{\rm b} = 0.1$, the two forms in Eq. (E1) agree to a maximum residual 5.61×10^{-14} at broadening 10^{-3}. This identity requires every anti-Hermitian part of the effective retarded kernel to have the matching lesser reservoir self-energy at the same distribution f. It does not hold for an arbitrary independently chosen non-Hermitian Hamiltonian. For the oscillatory protocol it is an assumed leading instantaneous adiabatic closure; for stationary rotation it requires consistent thermal preparation in that frame. Stationarity alone does not impose that preparation. The need for a complete doubled-contour construction is consistent with the exact high-bias Anderson-impurity solution of Oguri and Sakano [2], where a finite non-Hermitian Liouville–Fock Hamiltonian reproduces physical Keldysh Green functions only with the appropriate paired initial and final states. Their high-energy limit is cited as a structural precedent, not as a low-temperature approximation to the present driven kernel. In the diagonal-hybridization basis of $\tilde{H}$, the two channels $\tilde{d}_\pm$ are independently coupled to the $c_{k\pm}$ baths, and each satisfies the FDR: $[G^<_{\rm bath}]_{\pm\pm} = f(\omega)[G^A - G^R]_{\pm\pm}$.

b. Number continuity. The same complete self-energy satisfies the stationary continuity identity

$$\mathcal{I}_N = \int \frac{d\omega}{2\pi}\,\mathrm{Tr}\big[\Sigma^< G^> - \Sigma^> G^<\big] = 0. \tag{E2}$$

This check uses the full channel matrix, including the thermal partner of the relative width. It tests consistency of the assumed reservoir closure; it is not an independent demonstration of driven thermalization.

c. Physical DOS identity. For the complete physical kernel, $G^A = (G^R)^\dagger$ and

$$\mathcal{A} = i(G^R - G^A) = G^R \Gamma_{\rm tot} G^A, \qquad \Gamma_{\rm tot} \succeq 0. \tag{E3}$$

Consequently, $\rho_{\rm imp} = \frac{1}{2\pi}\mathrm{Tr}(P_{\rm imp}\mathcal{A}) = -\frac{1}{\pi}\mathrm{Im}\,\mathrm{Tr}(P_{\rm imp} G^R)$. The two expressions are identical, not competing density-of-states (DOS) prescriptions. We verify their equality and the positivity of the complete physical response numerically. A signed response of the compensated relative kernel after common damping is removed is only a kernel diagnostic and is not interpreted as a physical density of states.

Appendix F: Shifted Friedel-Type Resonance Condition

This appendix specifies the channel-diagonal SBMF resonance reference; it introduces no new Friedel sum rule.

1. Resonance Position in the Lorentzian Regime

For a channel-diagonal, constant-width reference with the coherent mixing set to zero ($D = 0$), the spinon retarded Green function is

$$G^R_{f,\sigma}(\omega) = \frac{1}{\omega - \tilde{\epsilon}_\xi + i\Gamma^{\rm net}_\sigma}, \qquad \Gamma^{\rm net}_\sigma = \Gamma_\sigma - x_\sigma, \tag{F1}$$

where $x_\sigma = \zeta_\sigma \Gamma_{\rm PT} = \pm\Gamma_{\rm PT}$ is the spin-selective imaginary shift from the coarse-grained drive and $\Gamma^{\rm net}_\sigma > 0$ in the stable low-energy regime. The corresponding coherent physical impurity spectral function, for $d = rf$, is

$$A_{d,\rm coh}(\omega) = \frac{r^2}{\pi} \sum_\sigma \frac{\Gamma^{\rm net}_\sigma}{(\omega - {\rm Re}\,\tilde{\epsilon}_\xi)^2 + (\Gamma^{\rm net}_\sigma)^2}. \tag{F2}$$

When the line shape is approximately Lorentzian, both spin channels are centred at the same reference frequency $\omega_{\rm res} \simeq {\rm Re}(\tilde{\epsilon}_\xi)$. The omitted incoherent spectral weight is outside this saddle description. For the EP-active matrix kernel the diagonal entries instead obey $G^R_{f,\uparrow\uparrow} = 1/(a - D^2/b)$ and $G^R_{f,\downarrow\downarrow} = 1/(b - D^2/a)$, with a, b defined in App. H; the scalar Lorentzians do not describe that problem. This resonance position is a channel-diagonal reference, not the eigenvalue of the independent 4×4 benchmark. A non-Lorentzian maximum can differ because of the line shape.

2. Origin of the Shift

In a conventional finite-U particle-hole-symmetric reference a resonance may be pinned by symmetry. The $U \to \infty$ saddle used here has no finite particle-hole-symmetric impurity level. Within the declared channel-diagonal reference the real shift is simply ${\rm Re}(\tilde{\epsilon}_\xi) = \epsilon_\xi + {\rm Re}(\delta_c) + \delta_\Sigma$. The shift is gauge invariant and independent of the phase of b_c.

a. Numerical reference. The value $\tilde{\epsilon}_\xi = -1$ in Fig. S1 is an assigned coefficient of an independent frozen benchmark, not a self-consistent local-moment result. The separate low-temperature Markov saddle is given in App. D 1.

Appendix G: Channel-Diagonal Slave-Boson Reference and Screening Scope

This appendix retains the channel-diagonal ($D = 0$) SBMF reference and separates its width scale from a conventional Schrieffer–Wolff estimate. The EP-active saddle is the matrix problem of App. D. No scalar replacement of that matrix problem or complex-level modulus in a Kondo exponent is used.

1. Effective Spinon Hamiltonian

For $d = rf$ the channel-diagonal reference is

$$H_{\rm eff} = \sum_\sigma \tilde{\epsilon}_\sigma f^\dagger_\sigma f_\sigma + \sum_{k\sigma} \big[r V_{k\sigma}\, c^\dagger_{k\sigma} f_\sigma + {\rm h.c.} \big], \tag{G1}$$

where the local imaginary term is a reduced-kernel parametrization of a separate passive reservoir, with its common loss and lesser partner retained; it is not generated again by the explicit bath in this reference. Here $V_{k\sigma}$ is the hybridization and $\tilde{\epsilon}_\sigma = \tilde{\epsilon}_\xi + i x_\sigma$ is the complex spinon level. Here $x_\sigma = \zeta_\sigma \Gamma_{\rm PT}$ parametrizes relative decay, and $\epsilon_{r,\sigma} = {\rm Re}\tilde{\epsilon}_\sigma = \tilde{\epsilon}_\xi$ is common to both spins. The chiral rotation is that of App. D. The renormalized hybridization width is $\Gamma_\sigma(\omega) = \pi r^2 \sum_k |V_{k\sigma}|^2 \delta(\omega - \varepsilon_{k\sigma}) \equiv r^2 \Gamma^{(0)}_\sigma(\omega)$, and the spinon retarded Green's function is $G^R_{f,\sigma}(\omega) = \frac{1}{\omega - \tilde{\epsilon}_\sigma + i\Gamma_\sigma}$.

2. Saddle-Point Equations at $T_{\rm b} = 0$

With the complex spinon level $\tilde{\epsilon}_\sigma = \epsilon_{r,\sigma} + i x_\sigma$ the net channel width is $\Gamma^{\rm net}_\sigma = \Gamma_\sigma - x_\sigma = r^2\gamma_\sigma$, with $\gamma_\sigma = \Gamma^{(0)}_\sigma - \zeta_\sigma \Gamma^{(0)}_{\rm PT}$. Stationarity of $\mathcal{F}_{\rm MF}$ with respect to δ_c gives the constraint, and stationarity with respect to r^2 gives the level equation. At $T_{\rm b} = 0$,

$$Q - r^2 = \sum_\sigma \int_{-D_{\rm uv}}^{0} \frac{d\omega}{\pi} \frac{\Gamma^{\rm net}_\sigma}{(\omega - \epsilon_{r,\sigma})^2 + (\Gamma^{\rm net}_\sigma)^2}, \tag{G2}$$

$$\delta_c = -\sum_\sigma \int_{-D_{\rm uv}}^{0} \frac{d\omega}{\pi} \frac{(\omega - \epsilon_{r,\sigma})\gamma_\sigma}{(\omega - \epsilon_{r,\sigma})^2 + (\Gamma^{\rm net}_\sigma)^2}. \tag{G3}$$

These scalar Read–Newns-type equations assume diagonal constant net widths scaling as r^2 and the stated thermal closure. They are not used for the EP-active numerical saddle, which retains coherent mixing.

3. Kondo-scale estimate near an EP

Within slave-boson mean-field theory the low-energy scale is set by the renormalized impurity width. With $r = |b_c|$ and $V_{\rm eff} = rV$, $\tilde{\Gamma}_\sigma(0) = r^2\Gamma^{(0)}_\sigma(0)$, where $\Gamma^{(0)}_\sigma(0) = \pi V^2 \rho_{b,\sigma}(0)$. This coherent width can be quoted as an SBMF reference scale. It does not derive an exponential screening law for a complex level. In particular, replacing a charge excitation energy by $|\tilde{\epsilon}_\sigma|$ in a Hermitian Kondo exponent is unsupported. The only screening estimate used for comparison is main Eq. (28), with positive real charge gaps and the appropriate bath density; it is not a result for the nearly empty full-momentum EP.

The complete resolvent and the finite-U charge sum contain no $1/s$ factor [main Eqs. (25)–(26)]; under reciprocal eigenvector rescaling every consistently transformed vertex–propagator–vertex product is invariant, so no Petermann, condition-number, or $Z^{\rm bio}$ multiplier is attached to $G^{R,A,<}$, a vertex, or the Kondo exponent. That identity does not remove genuine dependence carried by pole motion, matrix residues, interaction vertices, or relaxation rates; the finite-U matrix-chain and transition-residue calculations are independent consistency tests, not a second algebraic proof.

If $\Delta_{\rm eff} = D_0$ and $\Gamma_{\rm PT} = g_G\beta_0$, then $\partial_{\beta_0} s^2 = -2\Gamma_{\rm PT} g_G$. Both amplitudes can increase while s either increases or decreases. When s^2 increases, main Eq. (26) can increase through the ordinary charge denominators; this is not geometric amplification. For the pre-EP path used in the direct RG test, s^2 decreases toward zero; the resulting flow diagnostic is given in App. K.

Appendix H: Full Scattering Matrix and Contact Algebra

The Fisher–Lee and Wigner–Smith constructions used here are standard scattering tools [3–5]. Their role is to test causality and phase counting for the projected kernel; the novelty claimed in the article is the microscopic origin of that kernel.

1. Spin, chiral, and scattering-eigenchannel bases

Let $\Psi_s = (d_\uparrow, d_\downarrow)^T$ denote the original spin basis and $U_H = 2^{-1/2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$. For the compensated Markovian core,

$$\begin{aligned} h_s &= \tilde{\epsilon}_\xi I + D\sigma_x + iG\sigma_z - i\bar{\Gamma} I, \\ \Gamma_s &= i(\Sigma^R_s - \Sigma^A_s) = 2(\bar{\Gamma} I - G\sigma_z), \end{aligned} \tag{H1}$$

where $D = \Delta_{\rm eff}$ and $G = \Gamma_{\rm PT}$. With $z = \omega - \tilde{\epsilon}_\xi + i\bar{\Gamma}$ and $\mathcal{D}(z) = z^2 - D^2 + G^2$, direct inversion gives

$$G^R_s(\omega) = \frac{1}{\mathcal{D}(z)} \begin{pmatrix} z + iG & D \\ D & z - iG \end{pmatrix}. \tag{H2}$$

Writing $a = z - iG$ and $b = z + iG$ also gives the Schur forms

$$[G_s^R]_{\uparrow\uparrow} = \frac{1}{a - D^2/b}, \qquad [G_s^R]_{\downarrow\downarrow} = \frac{1}{b - D^2/a},$$
$$[G_s^R]_{\uparrow\downarrow} = [G_s^R]_{\downarrow\uparrow} = \frac{D}{ab - D^2}. \tag{H3}$$

Thus the local denominators are useful Schur complements, but they do not make the complete Green function or scattering matrix diagonal.

The fixed Hadamard/chiral fields are $\Psi_\chi = U_H^\dagger \Psi_s = (\widetilde{d}_+, \widetilde{d}_-)^T$. Every object, including the rate matrix, must be rotated:

$$h_\chi = U_H^\dagger h_s U_H = \tilde{\epsilon}_\xi I + D\sigma_z + iG\sigma_x - i\bar{\Gamma} I,$$
$$\Gamma_\chi = U_H^\dagger \Gamma_s U_H = 2(\bar{\Gamma} I - G\sigma_x),$$
$$G_\chi^R = U_H^\dagger G_s^R U_H = \frac{1}{\mathcal{D}(z)} \begin{pmatrix} z + D & iG \\ iG & z - D \end{pmatrix}. \tag{H4}$$

If $p = z - D$ and $q = z + D$, the corresponding chiral Schur coefficients are

$$[G_\chi^R]_{++} = \frac{1}{p + G^2/q}, \qquad [G_\chi^R]_{--} = \frac{1}{q + G^2/p},$$
$$[G_\chi^R]_{+-} = [G_\chi^R]_{-+} = \frac{iG}{pq + G^2}. \tag{H5}$$

Equations (H3) and (H5) are the corrected coefficient evaluation replacing the former block-diagonal contact argument.

2. Contact matching and Fisher–Lee coefficients

Choose an asymptotic-channel coupling matrix $\mathcal{W}_\nu$ such that $2\pi \mathcal{W}_\nu \mathcal{W}_\nu^\dagger = \Gamma_\nu$, with $\nu = s, \chi$. For an incoming channel vector $A_\nu^{\rm in}$, the impurity contact amplitude and outgoing vector satisfy

$$e_\nu(\omega) = G_\nu^R(\omega) \mathcal{W}_\nu A_\nu^{\rm in},$$
$$A_\nu^{\rm out} = A_\nu^{\rm in} - 2\pi i \mathcal{W}_\nu^\dagger e_\nu = S_\nu(\omega) A_\nu^{\rm in}. \tag{H6}$$

For the positive square-root factorization this gives

$$S_\nu = I - i\Gamma_\nu^{1/2} G_\nu^R \Gamma_\nu^{1/2}, \qquad S_\chi = U_H^\dagger S_s U_H. \tag{H7}$$

In the spin representation define $\gamma_\uparrow = 2(\bar{\Gamma} - G)$ and $\gamma_\downarrow = 2(\bar{\Gamma} + G)$. The explicit coefficients are

$$[S_s]_{\uparrow\uparrow} = 1 - i\gamma_\uparrow \frac{z + iG}{\mathcal{D}(z)},$$
$$[S_s]_{\downarrow\downarrow} = 1 - i\gamma_\downarrow \frac{z - iG}{\mathcal{D}(z)},$$
$$[S_s]_{\uparrow\downarrow} = [S_s]_{\downarrow\uparrow} = -i\sqrt{\gamma_\uparrow \gamma_\downarrow} \frac{D}{\mathcal{D}(z)}. \tag{H8}$$

The chiral coefficients must be obtained from the same covariant rotation, or equivalently from Eq. (H7) using Γ_χ and G_χ^R. It is incorrect to rotate G^R while leaving Γ diagonal.

For the validation point $\bar{\Gamma} = 0.12$ and $D = G = 0.10$, at $\omega = \tilde{\epsilon}_\xi$,

$$S_s = \begin{pmatrix} \mathsf{a} & i\mathsf{b} \\ i\mathsf{b} & \mathsf{a} \end{pmatrix},$$
$$S_\chi = \mathrm{diag}(\mathsf{a} + i\mathsf{b}, \mathsf{a} - i\mathsf{b}),$$
$$\mathsf{a} = 0.3888889, \qquad \mathsf{b} = 0.9212847. \tag{H9}$$

Consequently the earlier number 0.9213 is a spin-representation off-diagonal coefficient and, at the resonance center, the imaginary part of the two unit-modulus scattering eigenvalues. It is not a chiral conversion amplitude. In the same Markov gate, $[S_\chi(\tilde{\epsilon}_\xi)]_{+-} = 0$, while $\max_\omega |[S_\chi]_{+-}| = 0.8333333$ at $|\omega - \tilde{\epsilon}_\xi| = 0.12$. These component values are basis dependent; the eigenphases, $\det S$, and Wigner–Smith eigenvalues and trace are the quantities used in the basis-independent checks.

3. Four contact amplitudes and exact folded identity

The four coefficients that occur when the two incoming channels are treated simultaneously are the two columns of the impurity contact Green function. Let $e_{\eta'\eta}$ denote the impurity amplitude in fixed chiral component η' generated by an incoming asymptotic channel η, and let $\hat{e}_\eta$ be the unit vector of that channel. With $p = z - D, \quad q = z + D, \quad \mathcal{P}_0 = pq + G^2$, the exact quadratic contact equations are

$$\begin{pmatrix} p & -iG \\ -iG & q \end{pmatrix} \begin{pmatrix} e_{+\eta} \\ e_{-\eta} \end{pmatrix} = \mathcal{W}_\chi \hat{e}_\eta \Phi_\eta, \qquad \mathcal{W}_\chi = \left(\frac{\Gamma_\chi}{2\pi}\right)^{1/2}, \tag{H10}$$

where Φ_η is the incoming amplitude. Equation (H10) is the column form of Eqs. (H6) and (H4); it is exact for the quadratic projected kernel and introduces no two-particle assumption. Because $\Gamma_\chi = 2(\bar{\Gamma} I - G\sigma_x)$, the coupling $\mathcal{W}_\chi$ is not diagonal for $G \neq 0$, so each incoming channel feeds both impurity components; a scalar coupling would make S_χ nonunitary.

The Schur complements of the matrix on the left are the dressed contact denominators $\widehat{p} = p + G^2/q = \mathcal{P}_0/q$ and $\widehat{q} = q + G^2/p = \mathcal{P}_0/p$, with $[G^R_\chi]_{++} = 1/\widehat{p}$ and $[G^R_\chi]_{--} = 1/\widehat{q}$ [Eq. (H5)]. Consequently, $\frac{\widehat{p}}{\widehat{q}} = \frac{p}{q}$. This identity is a property of the diagonal elements of G^R_χ and holds for any incoming superposition. Both folded denominators contain the same quadratic pole polynomial $\mathcal{P}_0$; the ratio removes that common factor. It therefore exposes the common denominator responsible for the Jordan limit and for cancellation between individual projectors. It is not a basis-independent amplification factor, and amplitude ratios such as e_{--}/e_{++} depend on the incoming superposition through $\mathcal{W}_\chi$.

4. Controlled finite-U Anderson contact algebra

The four amplitudes in Eq. (H10) are the two columns of a *one-particle* Green function. They should not be confused with the two-electron exchange matrix. To obtain the latter, first impose the controlled reduction stated in the main text: equal folded velocities, linear dispersion, constant $M = D\sigma_z + iG\sigma_x$, momentum-independent scalar hybridization V, and the standard local interaction $Un_{d+}n_{d-}$. With momenta measured in energy units, the two-electron bath wave function in the ordered sector $x_1 < x_2$ is

$$\Psi_{ab}(x_1, x_2) = \mathcal{A}^{ab}_{12} e^{i(px_1 + qx_2)/v} + \mathcal{A}^{ab}_{21} e^{i(qx_1 + px_2)/v}, \qquad x_1 < x_2, \tag{H11}$$

supplemented by sectors with one bath electron and one impurity electron and by a doubly occupied impurity amplitude D_{pq}. Substitution in the Schrödinger equation, integration across the point contact, and elimination of those impurity amplitudes separate into triplet and singlet sectors. In the normalization $\Gamma_A = V^2/(2v)$, the resulting contact equations are [6–9]

$$\mathcal{A}^{(t)}_{21} = \mathcal{A}^{(t)}_{12}, \tag{H12}$$

$$[\Delta\mathcal{B} + i\,2U\Gamma_A]\,\mathcal{A}^{(s)}_{21} = [\Delta\mathcal{B} - i\,2U\Gamma_A]\,\mathcal{A}^{(s)}_{12}, \tag{H13}$$

$$\Delta\mathcal{B} = \mathcal{B}(p) - \mathcal{B}(q),$$

$$\mathcal{B}(p) = p(p - 2\epsilon_d - U). \tag{H14}$$

Equations (H12) and (H13) display exactly where the virtual doublon enters: it changes only the antisymmetric pseudospin channel. With $\Pi_t = (I + P_{12})/2$ and $\Pi_s = (I - P_{12})/2$,

$$\begin{aligned} \mathcal{A}_{21} &= R_{12}(p,q)\mathcal{A}_{12}, \\ R_{12}(p,q) &= \Pi_t + \frac{\Delta\mathcal{B} - i\,2U\Gamma_A}{\Delta\mathcal{B} + i\,2U\Gamma_A}\Pi_s \\ &= \frac{\Delta\mathcal{B}\, I_{12} + i\,2U\Gamma_A P_{12}}{\Delta\mathcal{B} + i\,2U\Gamma_A}. \end{aligned} \tag{H15}$$

Thus the contact matrix follows directly from the two projector-resolved matching equations; no rank-one Feshbach ansatz is added.

For $U \neq 0$, set $u(p) = \frac{\mathcal{B}(p)}{2U\Gamma_A}, \qquad x = u(p) - u(q)$. Then

$$R_{12}(x) = \frac{xI_{12} + iP_{12}}{x + i} = \frac{1}{x+i} \begin{pmatrix} x+i & 0 & 0 & 0 \\ 0 & x & i & 0 \\ 0 & i & x & 0 \\ 0 & 0 & 0 & x+i \end{pmatrix}. \tag{H16}$$

The bare pair-energy factor is absorbed exactly because $\mathcal{B}(p)-\mathcal{B}(q)=(p-q)(p+q-2\epsilon_d-U)$. The rapidity map is generically two-to-one, $u(p)=u(2\epsilon_d+U-p)$, and is singular at $U=0$. Global injectivity is not required for the matrix algebra, while the $U=0$ quadratic benchmark is treated separately.

For the unnormalized matrix $\mathcal{R}_{12}(x)=xI_{12}+iP_{12}$, the permutation identities give

$$\mathcal{R}_{12}(u-v)\mathcal{R}_{13}(u-w)\mathcal{R}_{23}(v-w)=\mathcal{R}_{23}(v-w)\mathcal{R}_{13}(u-w)\mathcal{R}_{12}(u-v), \tag{H17}$$

and, for every $S\in GL(2,\mathbb{C})$,

$$(S\otimes S)^{-1}\mathcal{R}_{12}(x)(S\otimes S)=\mathcal{R}_{12}(x), \tag{H18}$$

because $[P_{12},S\otimes S]=0$. For $S=S_C=e^{\theta\sigma_y/2}$, $\tanh\theta=G/D$,

$$S_C^{-1}(D\sigma_z+iG\sigma_x)S_C=\sqrt{D^2-G^2}\,\sigma_z. \tag{H19}$$

The same canonical transformation on bath and impurity fields leaves the kinetic term, scalar hybridization, and $n_{d+}n_{d-}=N_d(N_d-1)/2$ invariant. It therefore preserves the ordinary YBE rather than producing a new bulk R-matrix.

The non-diagonal pseudospin sector can be displayed without choosing the diagonal positive-metric frame. Define

$$\mathcal{J}^\mu=\frac{1}{2}\int_0^{\mathcal{L}}dx\,\bar\chi\sigma_\mu\chi+\frac{1}{2}\bar d\sigma_\mu d,\qquad \mathcal{J}^\pm=\mathcal{J}^x\pm i\mathcal{J}^y. \tag{H20}$$

For $M=D\sigma_z+iG\sigma_x$, the charge conserved by the spin-invariant reference Anderson Hamiltonian is

$$Q_M=2D\mathcal{J}^z+iG(\mathcal{J}^+-\mathcal{J}^-). \tag{H21}$$

Adding this total charge defines a comparison family in which the matrix acts on both bath and impurity. A bath-only deformation instead contains $Q_{b,M}=Q_M-\bar dMd$; hybridization does not conserve $Q_{b,M}$. These are distinct operators and boundary problems. In both cases the nonzero ladder terms (often written $S^\pm$) are retained. For $D^2\neq G^2$, the global canonical action induced by S_C maps Eq. (H21) to $2\sqrt{D^2-G^2}\,\mathcal{J}^z$, while Eq. (H18) keeps the same Yang–Baxter algebra.

At $D^2=G^2$, the diagonalizer in Eq. (H19) is singular, but $M^2=0$. On a finite ring the position-dependent gauge map can remove the constant bath matrix from the first-order differential expression, producing the seam holonomy

$$\mathcal{G}=e^{iM\mathcal{L}/v}=I+iM\mathcal{L}/v,\qquad \mathcal{G}^{-1}=I-iM\mathcal{L}/v. \tag{H22}$$

The transformed impurity/contact terms and the boundary domain must also be retained. This finite-ring construction does not imply a uniformly bounded infinite-line similarity, nor does it apply to an arbitrary local impurity matrix in a fixed undeformed bath. Because Eq. (H18) also holds for this non-diagonalizable $\mathcal{G}$, the monodromy

$$T_a(z)=\mathcal{G}_a\mathcal{R}_{aN}(z-u_N)\cdots\mathcal{R}_{a1}(z-u_1) \tag{H23}$$

satisfies twisted RTT and generates commuting transfer matrices. The finite-volume domain construction and local descendant-root confluence are developed in the revised companion work [10]. Its bath-only Jordan theorem assumes an isolated eigenvalue whose complete eigenspace is one spin-S multiplet and a nonzero projected bath-spin coefficient. It follows the continued spectral subspace, since that deformation does not preserve the original eigenspace. The conserved-total-charge family is a separate comparison. Neither theorem establishes a resonance EP of the frequency-dependent kernel studied here, or its survival under general interaction self-energies.

The supplied script `verify_finiteU_contact_algebra.py` independently checks the displayed 4×4 matrix. For generic complex rapidities and a generic non-unitary S, it gives Frobenius residuals 1.14×10^{-15} for YBE, 4.48×10^{-16} for $GL(2,\mathbb{C})$ covariance, and 1.66×10^{-16} for the metric diagonalization; the singlet and triplet eigenvalue residuals vanish to machine precision.

5. One-particle contact-phase quantization

Inside the passivity window, $S^\dagger S=I$. Let $\mathtt{s}_a=e^{2i\delta_a}$, $a=1,2$, denote its two eigenvalues. These scattering eigenchannels are not identified with the fixed chiral labels. Periodic closure on a ring after the stated finite-window radial linearization gives the exact one-particle phase condition

$$\det\bigl[I-e^{ik\mathcal{L}}S(k)\bigr]=0\quad\Longleftrightarrow\quad e^{ik_{ja}\mathcal{L}}\mathtt{s}_a(k_{ja})=1. \tag{H24}$$

This is exact phase counting for the complete causal one-particle matrix. It is logically distinct from the multiparticle contact matrix Eq. (H15), which requires the additional controlled linearization stated above.

Appendix I: Biorthogonal Structure and Eigenvectors

The frozen impurity block is $h_d = \left[\begin{smallmatrix} \tilde{\epsilon}_\xi + D & iG \\ iG & \tilde{\epsilon}_\xi - D \end{smallmatrix}\right]$ with splitting $s = \sqrt{D^2 - G^2}$ and eigenvalues $E_\pm = \tilde{\epsilon}_\xi \pm s$.

a. Right and left eigenvectors. Choose $|r_\pm\rangle = [G, i(D \mp s)]^T$. Since h_d is complex symmetric, the left rows can be chosen as $\langle l_\pm| = [G, i(D \mp s)]$, without complex conjugation. Then $\langle l_\pm | r_\mp \rangle = 0$, $\langle l_+ | r_+ \rangle = 2s(D - s)$ and $\langle l_- | r_- \rangle = -2s(D + s)$. The product of left and right normalization factors is therefore $\mathcal{M}_\pm \mathcal{N}_\pm = 1/\langle l_\pm | r_\pm \rangle \sim 1/s$ near a nonzero EP. This singular normalization is not an additional physical spectral weight: reciprocal eigenvector rescalings leave each normalized projector unchanged, and the divergent parts of the two projectors cancel in the complete resolvent [main Eq. (25)]. This agrees with $\kappa_{\rm imp} \sim C/|s|$ [Eq. (A1)]. At the EP the right eigenvectors coalesce, and a generalized vector satisfies $(h_d - E_{\rm EP})|j\rangle = |r_+\rangle$, with $h_d |r_+\rangle = E_{\rm EP} |r_+\rangle$.

1. Explicit Contact Projector

The normalized rank-one projector in the $\{\tilde{d}_+, \tilde{d}_-\}$ contact space is

$$P_+ = \frac{|r_+\rangle\langle l_+|}{\langle l_+ | r_+ \rangle} = \frac{1}{2s(D-s)} \begin{pmatrix} G^2 & iG(D-s) \\ iG(D-s) & -(D-s)^2 \end{pmatrix}, \tag{I1}$$

with $P_+^2 = P_+$. The full Hamiltonian $\tilde{H}(k)$ contains additional diagonal bath dispersions $\varepsilon_{c,\pm}(k)$ that do not enter this local projector. The projector is used only to track biorthogonal normalization; by itself it does not establish a two-body R-matrix for the full energy-dependent channel-converting problem. The distinct finite-U contact derivation under controlled linearization is given in Sec. H 4.

Appendix J: Causal Fisher–Lee and Wigner–Smith Check

For the compensated Markovian small-width kernel in the spin representation, set $H_s = D\sigma_x - i\bar{\Gamma} I + iG\sigma_z$ and $G_s^R = (\omega I - H_s)^{-1}$. Passivity, $\bar{\Gamma} \geq |G|$, makes $\Gamma_s = 2(\bar{\Gamma} I - G\sigma_z)$ positive semidefinite. The scattering and delay matrices are $S_s = I - i\Gamma_s^{1/2} G_s^R \Gamma_s^{1/2}$ and $Q_s = -iS_s^\dagger \partial_\omega S_s$. The chiral matrices follow by the simultaneous rotation in Eq. (H7). Direct use of the Dyson identity gives $S_\nu^\dagger S_\nu = I$ and $Q_\nu = Q_\nu^\dagger$ in either representation $\nu = s, \chi$. The trace also obeys

$$\mathrm{Tr}\, Q_\nu = -i\, \partial_\omega \log \det S_\nu. \tag{J1}$$

In the compensated unbroken Markov core, let $x = \omega - \tilde{\epsilon}_\xi$ and $s = \sqrt{D^2 - G^2} \in \mathbb{R}$. Causality fixes the scattering determinant entirely from the two poles and their advanced zeros:

$$\det S(\omega) = \prod_{\alpha=\pm} \frac{x - \alpha s - i\bar{\Gamma}}{x - \alpha s + i\bar{\Gamma}}, \tag{J2}$$

$$\mathrm{Tr}\, Q(\omega) = \sum_{\alpha=\pm} \frac{2\bar{\Gamma}}{(x - \alpha s)^2 + \bar{\Gamma}^2}. \tag{J3}$$

At the resonance center this gives the analytic pre-EP concentration $\mathrm{Tr}\, Q(\tilde{\epsilon}_\xi) = \frac{4\bar{\Gamma}}{s^2 + \bar{\Gamma}^2}$. Relative to the matched Hermitian kernel,

$$\frac{\mathrm{Tr}\, Q_G(\tilde{\epsilon}_\xi)}{\mathrm{Tr}\, Q_{G=0}(\tilde{\epsilon}_\xi)} = \frac{D^2 + \bar{\Gamma}^2}{D^2 - G^2 + \bar{\Gamma}^2} > 1 \tag{J4}$$

for $0 < |G| < |D|$ at fixed D and $\bar{\Gamma}$. Resolving the determinant into the two eigenvalues s_a of the complete unitary matrix gives

$$\frac{1}{2\pi} \mathrm{Tr}\, Q_\nu = \sum_{a=1}^{2} \frac{1}{2\pi i} \partial_\omega \log \mathsf{s}_a \equiv K_{\rm ph}. \tag{J5}$$

Thus the logarithmic contact-phase density of Eq. (27) is not an independent scale ansatz: it is the causal scattering density-of-states shift of the same complete retarded/advanced problem. At $\beta_0 = 0.50$, the independent full-matrix

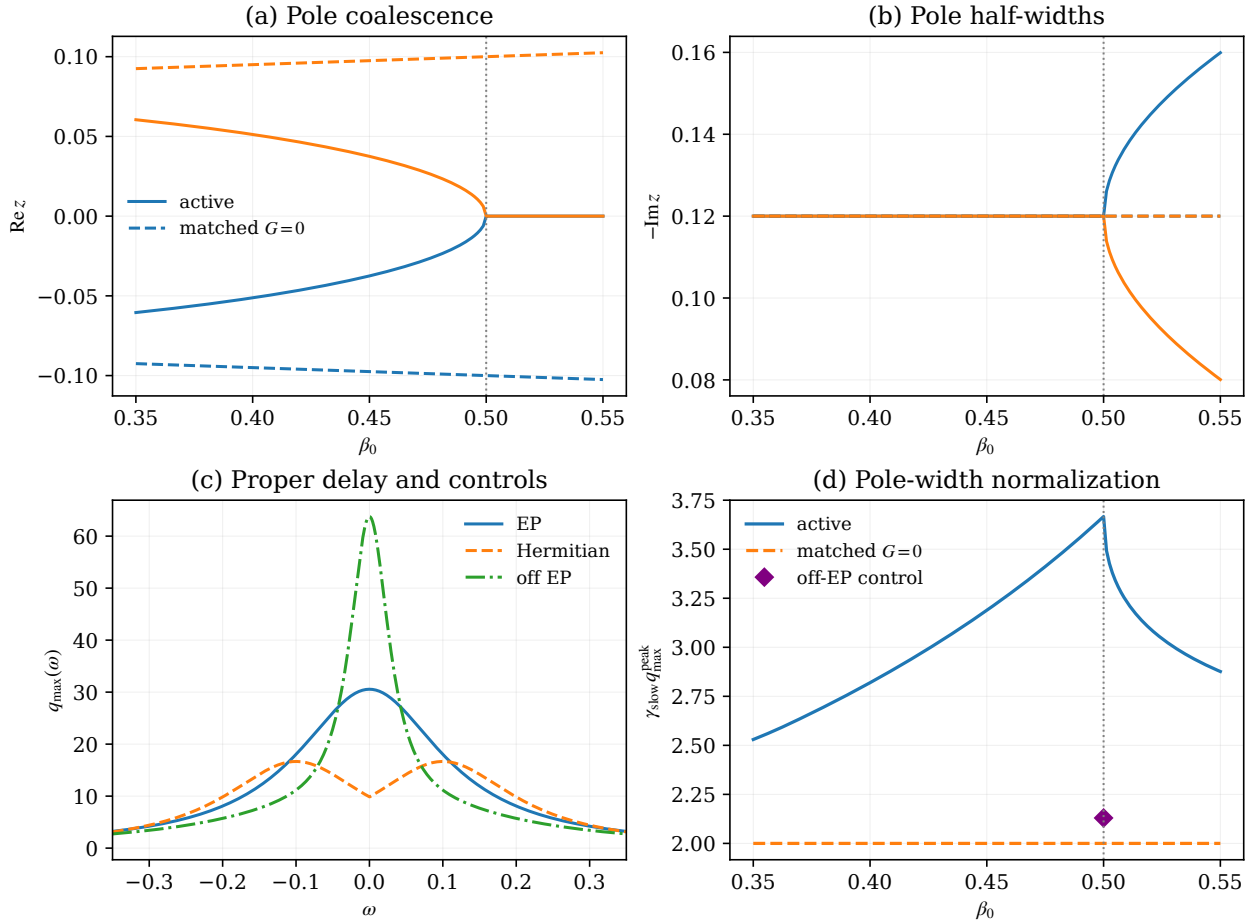


FIG. S2. **Causal Smith-delay validation and adverse controls.** Panel (a) tracks the analytic pole pair to the exact core EP at $\beta_0 = 0.50$. Panel (b) shows the two pole half-widths: they remain positive throughout the retained window and split only on the relative-broken side. Panel (c) compares the largest proper-delay peak with a Hermitian control matched by total rate and with an off-EP kernel of the same total rate ($D = 0.05$); the latter exceeds the EP peak because one of its poles decays at only 0.0334. Panel (d) multiplies the largest proper delay by the slowest pole half-width $\gamma_{\rm slow} = \min_a[-\operatorname{Im} z_a]$. The separate off-EP control is marked at $\beta_0 = 0.50$. The passivity floor $\bar{\Gamma} - |\Gamma_{\rm PT}|$ is not used as a pole width. Together with the unitarity and trace-delay checks, these controls establish the proper delay as a causal coalescence and linewidth fingerprint, not a Kondo-scale multiplier.

gate gives a maximum scattering-eigenvalue modulus error 9.99×10^{-16} and a finite-difference phase–Smith identity error 5.53×10^{-9}. The basis-resolved coefficients are given in Eq. (H9); no individual off-diagonal matrix element is used as a basis-independent diagnostic.

At the EP, $D = G$ and the pole pair coalesces at $E_{\rm EP} = -i\bar{\Gamma}$. The complete Jordan resolvent gives

$$\operatorname{Tr} Q_{\rm EP}(\omega) = \frac{4\bar{\Gamma}}{\omega^2 + \bar{\Gamma}^2}, \qquad \int \frac{d\omega}{2\pi} \operatorname{Tr} Q_{\rm EP} = 2. \tag{J6}$$

For $\bar{\Gamma} = 0.12$, this predicts $\operatorname{Tr} Q_{\rm EP}(0) = 33.3333333$, reproduced numerically to machine precision. The finite-window numerical integral is 1.99236; the small deficit is the omitted Lorentzian tail. Across the validation grid, the maximum scattering-unitarity error is 1.33×10^{-15}, the maximum Smith-Hermiticity error is 2.52×10^{-14}, and the maximum discrepancy in Eq. (J1) is 4.97×10^{-14}.

Figure S2 includes the essential adverse control. An off-EP kernel with the same total rate and $D = 0.05$ reaches a largest proper-delay peak 63.77, exceeding the EP value 30.56, because one of its poles decays at only 0.0334. Normalized by the slowest pole half-width (both EP poles have half-width 0.12), the largest proper delay is 3.667 at the EP, 2.000 for the Hermitian control, and 2.130 for the off-EP kernel, so the EP value is 1.83 and 1.72 times larger. Figure S2(d) uses these pole half-widths consistently. The trace delay, by contrast, is the same at the EP and at the Hermitian degeneracy, $\operatorname{Tr} Q(0) = 4/\bar{\Gamma} = 33.33$, because it depends only on the pole positions. The proper delay

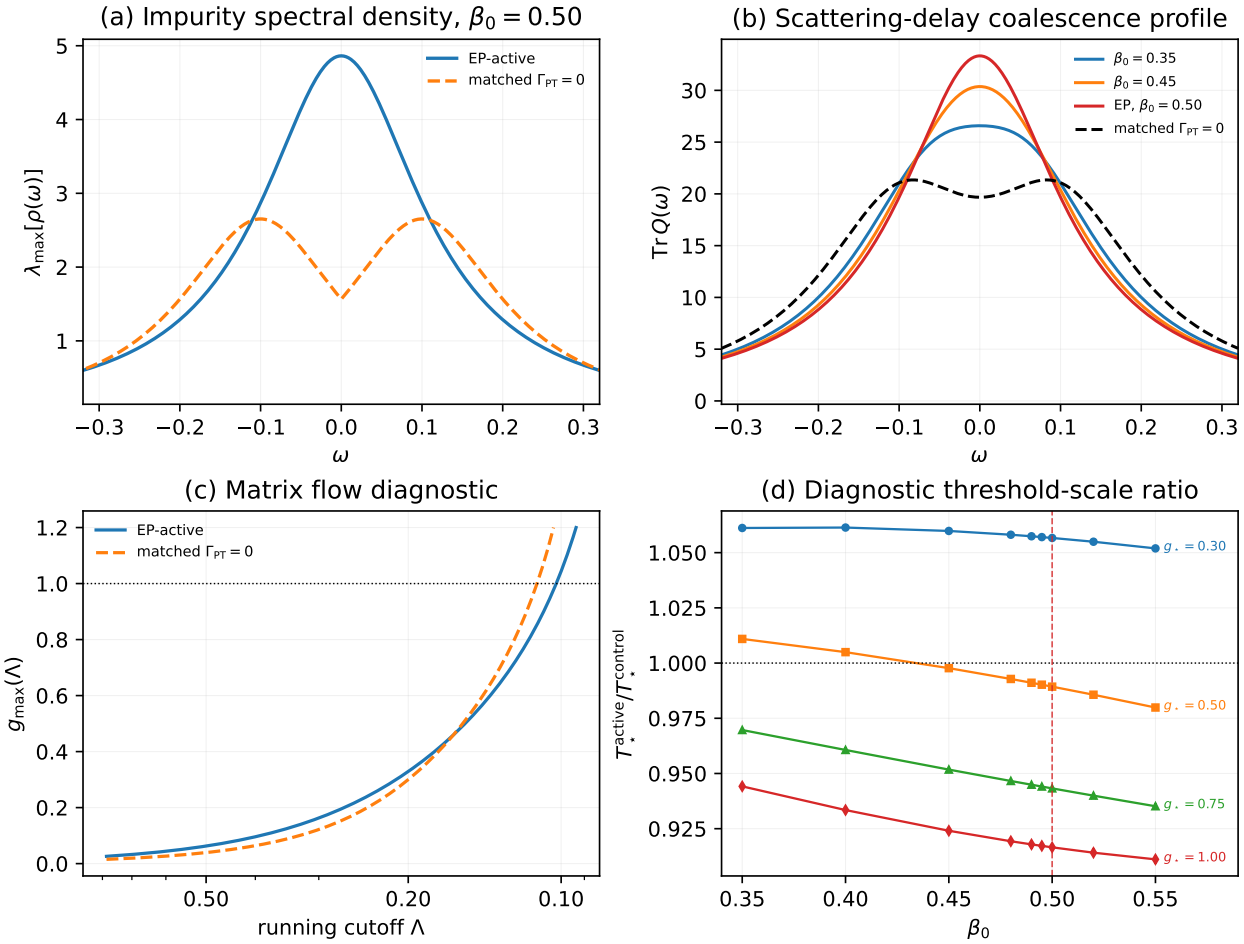


FIG. S3. **Causal scattering and spectral-density flow diagnostic.** The fixed-W validation uses the positive impurity spectral matrix built from G^R and G^A as the channel density (App. K), with $\Gamma_0 = 0.12$ and unit charge gap. Panel (a) compares the largest causal channel-density eigenvalue at $\beta_0 = 0.50$ with the matched $\Gamma_{\mathrm{PT}} = 0$ control. Panel (b) gives the Wigner–Smith trace; the passive Fisher–Lee matrix is unitary to 1.33×10^{-15} and $\mathrm{Tr}\,Q(0) = 4/\Gamma_0 = 33.3333$ at the EP. Panels (c) and (d) show that the EP-active exchange eigenvalue is initially larger, but its threshold-scale ratio crosses below the control before strong coupling. This supports a bounded early-flow ordering of this diagnostic, not an enhanced thermodynamic Kondo temperature.

therefore resolves the coalescence; it is used as a causal coalescence and linewidth fingerprint, not as an independent many-body amplification factor.

Appendix K: Causal Spectral Density and a Matrix Flow Diagnostic

This appendix records the complete Green-function matrix analysis used for Supplemental Fig. S3. It applies the one-loop matrix recursion patterned on poor-man's scaling [11] to the positive impurity spectral matrix. That matrix is not the bath spin density entering a derived Kondo RG equation. The calculation is an operational spectral-weight diagnostic throughout this appendix.

1. Positive matrix density

For either the spin or chiral representation,

$$\begin{aligned}\rho_\nu(\omega) &= \frac{i}{2\pi}\left[G^R_\nu(\omega) - G^A_\nu(\omega)\right] \\ &= \frac{1}{2\pi} G^R_\nu(\omega)\Gamma_\nu(\omega)G^A_\nu(\omega), \qquad \nu = s, \chi. \end{aligned} \tag{K1}$$

The second equality is the retarded/advanced Dyson identity. It proves positivity directly: $v^\dagger \rho_\nu v = \frac{1}{2\pi}\left\|\Gamma^{1/2}_\nu G^A_\nu v\right\|^2 \geq 0$. It also makes basis covariance explicit, $\rho_\chi = U_H^\dagger \rho_s U_H$. Consequently the two density eigenvalues are independent of whether the calculation is displayed in the spin or fixed chiral basis.

2. Exchange matrix and flow

We initialize the diagnostic with the conventional scalar Schrieffer–Wolff reference value,

$$J_0 = 2W^2\left(\frac{1}{|\epsilon_d|} + \frac{1}{|\epsilon_d + U|}\right) I. \tag{K2}$$

Let $\ell = \ln(D_{\rm uv}/\Lambda)$ and define the symmetric shell density $\rho_{\rm sh}(\Lambda) = \rho(+\Lambda) + \rho(-\Lambda)$. The matrix recursion integrated in this diagnostic is

$$\frac{dJ}{d\ell} = J(\ell)\rho_{\rm sh}(\Lambda)J(\ell). \tag{K3}$$

No simultaneous diagonalization of J and $\rho_{\rm sh}$ is assumed. For Hermitian J and positive Hermitian $\rho_{\rm sh}$, $J\rho_{\rm sh}J$ is Hermitian, so the flow preserves Hermiticity. Under a fixed unitary basis change, $J \mapsto U^\dagger J U$ and $\rho \mapsto U^\dagger \rho U$, making Eq. (K3) covariant.

To compare channels without assigning significance to a basis component, the code forms $\rho_{\rm av} = \rho_{\rm sh}/2$ and $g = \rho_{\rm av}^{1/2} J \rho_{\rm av}^{1/2}$, and follows $g_{\max} = \lambda_{\max}(g)$. The operational scale $T_*(g_*)$ is the cutoff at which $g_{\max}(\Lambda)$ first reaches a specified threshold g_*. It is deliberately called an operational weak-flow scale, not a thermodynamic Kondo temperature.

3. Coefficient and convergence audit

For $k_{\max} = \pi/4$ the gate uses the one-dimensional normalization $\rho_{\rm ref} = 1/(2k_{\max}) = 0.637$ (the radial density of the quadratic bath would be $1/(2\pi\alpha_{\rm b}) = 0.159$), $\Gamma_0 = \pi\rho_{\rm ref}W^2 = 0.12$, and therefore, with $\epsilon_d = -1$ and $U = 2$ in Eq. (K2), $W^2 = 0.06$ and $J_0 = 0.24$. At $\beta_0 = 0.50$, $D_0 = G_0 = 0.10$ and the two rate eigenvalues are 0.04 and 0.44. The selected-grid minimum density eigenvalue is 0.0365034, while the strict pre-EP-window minimum is $0.0241015 > 0$. The maximum difference between the two forms of Eq. (K1) is 1.79×10^{-15}, the exchange Hermiticity error is zero, and the maximum step-refinement drift is 5.92×10^{-10}.

The active/control threshold-scale ratios at $\beta_0 = 0.50$ are 1.0567, 1.0179, 0.9893, and 0.9166 for $g_* = 0.30$, 0.40, 0.50, and 1.00, respectively. The ordering therefore reverses near $g_* = 0.4627$. The calculation establishes a bounded early-flow ordering, but not a threshold-independent enhancement of the strong-coupling Kondo scale. In this flow ρ_ν of Eq. (K1) is the impurity spectral matrix of the projected kernel, used in place of the bath spin density at the impurity site that enters a poor-man's-scaling equation. Because $J_0 \propto I$ [Eq. (K2)], every EP dependence of the flow, including these ratios, enters through that substitution. The ratios are therefore an operational diagnostic of the impurity resolvent; a Kondo-flow statement would require the bath spin density.

Appendix L: Finite-U Biorthogonal NRG Residue Audit

We next test the Jordan response by an operator-resolved non-Hermitian NRG calculation in the scalar-hybridization Wilson-chain control of the projected model [12]. This is a weak-coupling structural control of the local finite-U residue in the early Wilson shells, not a low-energy fixed point or a phase diagram for the full SOC-resolved bath. It

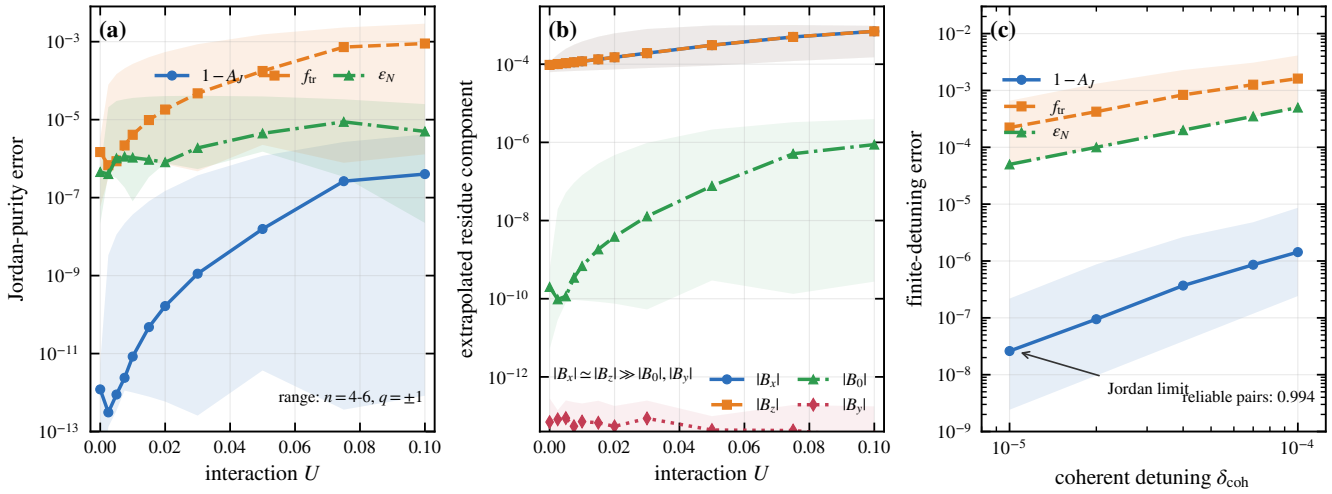


FIG. S4. **Finite-U Jordan transition-residue audit for the scalar-hybridization NRG control.** (a) Zero-detuning extrapolations of the alignment error $1-A_J$, trace fraction $f_{\rm tr}$, and nilpotent mismatch ϵ_N. Curves are medians and bands are the full range over the early/high-energy iterations $n=4$–6 and charge sectors $q=\pm1$. (b) Pauli components of the extrapolated double-pole coefficient B: $|B_x|\simeq|B_z|\gg|B_0|,|B_y|$, as required by $N_{\rm EP}=\Delta_{\rm eff}\sigma_z+i\Gamma_{\rm PT}\sigma_x$. (c) Direct finite-detuning convergence; curves are medians and bands are the 16th–84th percentile over U, iteration, and tracked sector. The reliable-pair fraction is 0.994. Parameters are $\Lambda=3$, $z=0.5$, $N_{\rm keep}=320$, $r=1$, $\beta_0=0.50$, $\Delta_{\rm eff}=\Gamma_{\rm PT}=0.10$, and $0\le U\le0.10$. The figure establishes persistence of the nilpotent matrix-residue direction in this weak-coupling structural window; it does not establish a low-energy fixed point, pole coalescence, a universal interaction law, a thermodynamic Kondo scale, or a phase boundary.

therefore neither locates nor excludes the interacting continuation of the full-k causal pole EP. That distinct problem is addressed by the exact finite-chain matrix block-Wilson calculation of Sec. C and by the continuum saddle of App. N. The present section instead asks whether the scalar Markov-core transition residue retains the nilpotent matrix direction. The two transition operators are propagated independently as $L^\dagger cR$ and $L^\dagger c^\dagger R$; a no-truncation regression checks their noninteracting matrix elements before the interacting scan.

At the exact Jordan point a complete biorthogonal eigenbasis is singular. We therefore calculate at finite coherent detuning $\delta_{\rm coh}>0$, track the same two impurity-supported poles through U and $\delta_{\rm coh}$, and only then extrapolate to $\delta_{\rm coh}\to0$. If

$$G(z)=\frac{W_+}{z-z_+}+\frac{W_-}{z-z_-}+G_{\rm reg}(z), \tag{L1}$$

the coefficient that approaches the Jordan double-pole residue is

$$B=\frac{1}{2}(W_+-W_-)(z_+-z_-)=B_0I+B_x\sigma_x+B_y\sigma_y+B_z\sigma_z. \tag{L2}$$

In the Hadamard/chiral basis used by the Wilson-chain adapter, the compensated EP generator is $N_{\rm EP}=\Delta_{\rm eff}\sigma_z+i\Gamma_{\rm PT}\sigma_x$, with $N_{\rm EP}^2=0$. We monitor the Frobenius-angle alignment, normalized trace, and nilpotent component mismatch,

$$A_J=\frac{|\operatorname{Tr}(N_{\rm EP}^\dagger B)|}{\|N_{\rm EP}\|_F\|B\|_F},$$
$$f_{\rm tr}=\frac{|\operatorname{Tr}B|}{\sqrt{2}\|B\|_F},\qquad \epsilon_N=\frac{|B_x-iB_z|}{|B_x|+|B_z|}. \tag{L3}$$

Supplemental Fig. S4 reports the validated part of the audit. The reference calculation uses $\Lambda=3$, $z=0.5$, $N_{\rm keep}=320$, bath exponent $r=1$, $\beta_0=0.50$, $\Delta_{\rm eff}=\Gamma_{\rm PT}=0.10$, $0\le U\le0.10$, and the early/high-energy NRG iterations $n=4$–6 in charge sectors $q=\pm1$. Five detunings between 10^{-5} and 10^{-4} are used. The reliable tracked-pair fraction is 0.993939. Across 328 reliable direct finite-detuning rows the median alignment is 0.9999997628, the median trace fraction is 5.77076387×10^{-4} and the median mismatch is 1.99999992×10^{-4}. The separate ensemble of 326 extrapolation fits has median extrapolated mismatch 7.31791124×10^{-7}; its medians of input-window alignment

and trace-fraction medians are 0.9999996308 and $8.35670512 \times 10^{-4}$. These are different statistical populations. Both support a nearly traceless nilpotent residue direction within the tested window.

Symmetry content. In this control the bath and $Un_\uparrow n_\downarrow$ are spin-rotation invariant, and the core is a complex impurity field $\boldsymbol{B} \cdot \boldsymbol{\sigma}$. Complex-spin covariance constrains the characteristic polynomial through $\boldsymbol{B} \cdot \boldsymbol{B}$; its null-field condition remains $\Delta_{\text{eff}} = \Gamma_{\text{PT}}$. Where the response is analytic, the Green function at that condition has the form $g_0 I + g_1 N_{\text{EP}}$. Covariance fixes the direction of a nonzero nilpotent component, but does not alone prove a nonzero double-pole residue of a selected resonance for every U. In particular, normalized alignment can approach one even if the residue amplitude tends to zero. The measured alignment therefore checks the transition-operator calculation in the stated window; it is not a theorem of interaction-protected pole coalescence. Three-site exact diagonalization at $U = 8$ checks spectral covariance against a field $(\boldsymbol{B} \cdot \boldsymbol{B})^{1/2}\sigma_z$ to 10^{-13} in the three-, four- and five-electron sectors. The spin-asymmetric full-k bath is instead treated in Apps. C and N.

1. An exact interacting comparison with a scalar bath

To distinguish symmetry-fixed direction from a nonzero residue, consider a separate finite Anderson control with spin-independent bath hopping and hybridization, and an imposed local field $N = \Delta\sigma_x + i\gamma\sigma_z$. It is not a replacement for the stationary SOC bath: making that bath scalar would remove the relative-decay mechanism, and inserting N on the impurity changes the model. Related interaction-stable symmetry-protected EPs are known [13].

For an isolated singlet of the Hermitian reference, define the response by biorthogonal analytic continuation of that state. Complex-spin covariance gives $G(z) = g_0(z)I + g_1(z)N$ at $\gamma = \Delta > 0$, where $N^2 = 0$. For an isolated spin-doublet transition of energy ω_ℓ, let Z_ℓ be its per-spin spectral weight and $m_\ell = \partial_h \omega_{\ell,\uparrow}(h)|_{h=0}$ its response to a local real field $h(n_{d\uparrow} - n_{d\downarrow})$. Differentiating the Hermitian spin-resolved pole gives the double-pole coefficient $B_\ell = Z_\ell m_\ell N$. Thus nonzero $Z_\ell m_\ell$, not normalized alignment alone, supplies the missing condition. For $K = G^{-1}$, $K(\omega_\ell) = -(m_\ell/Z_\ell)N$ has rank one and $\det K = (z - \omega_\ell)^2/Z_\ell^2 + O((z - \omega_\ell)^3)$.

For a half-filled dimer with one bath site at zero energy, hopping $V > 0$ and $\epsilon_d = -U/2$, define $a = \sqrt{U^2/16 + V^2}$, $c = \sqrt{U^2/16 + 4V^2}$ and $\omega_{\ell,h} = c \mp a$. Direct diagonalization gives

$$Z_\ell = \frac{\omega_\ell^2 - 9V^2}{2(\omega_\ell^2 - \omega_h^2)}, \qquad m_\ell = \frac{1}{2}\left(1 + \frac{U}{4a}\right). \tag{L4}$$

Both are positive for every finite $U \geq 0$. The scalar Green function is $g_0 = [z - V^2/z - (U^2/4)z/(z^2 - 9V^2)]^{-1}$, so a dynamical interaction self-energy is retained. Its nilpotent component is also nonzero in general: $\Sigma_1 = g_1/g_0^2 - 1$, rather than a frozen scalar ansatz. The reference has $n_d = 1$ and $D_d = [1 - U/\sqrt{U^2 + 64V^2}]/4$. The double pole persists at every finite U, although $Z_\ell m_\ell \sim 18V^2/U^2$ tends to zero at fixed V.

Fourteen untruncated checks cover one and three bath sites at $U = 0, 0.5, 2, 4, 8, 16$ and five sites at $U = 4, 8$, with all hoppings 0.5 and $\Delta = \gamma = 0.1$. Direct contour residues agree with $Z_\ell m_\ell$ within 2.1×10^{-15}; the selected charge-sector nullities of $H - E$ and $(H - E)^2$ are one and two. For five bath sites at $U = 8$, $n_d = 1$, $D_d = 0.01603$ and $Z_\ell m_\ell = 0.02548$. Two-sided splitting and pole-exchange loops are verified at $U = 4, 8$ with three bath sites. Here internal PT means spin exchange on every site followed by complex conjugation. A real bath σ_x field preserves it but splits the poles at the old null-field condition, showing that PT alone does not pin the EP. These are finite-system, biorthogonal response statements; no passive microscopic realization, continuum resonance or Kondo screening follows.

a. Limits of the original NRG scan. The stronger pole-splitting claims do not pass the same convergence gate. Across iteration, sector, fitting window, and detuning model, the complex interaction-exponent ensemble has median $p = 1.359$ with a 68% spread of 1.991. The proposed complex quartic coefficient also fails its plateau test (median absolute log-slope 0.880 and relative complex scatter 1.118). These failures do not prove the absence of a scaling regime, but they preclude extracting a universal exponent, quartic coefficient, or interaction-induced EP floor from this calculation. Accordingly only the matrix-residue result is used in the main text.

Appendix M: Scope and Division of Labor

Four statements are kept separate. (i) For the complete frozen retarded kernel the exact claims are the matrix Fisher–Lee relation, one-particle contact-phase quantization, and the Wigner–Smith identity; they require no many-body factorization. (ii) Only after equal-velocity branchwise linearization and folding, with constant M, scalar hybridization, and component-symmetric local U, is the two-electron contact matrix Eq. (H15); its difference form and $GL(2, \mathbb{C})$ covariance give the YBE, and, under the companion finite-ring contact and boundary assumptions, the

global nilpotent deformation retains its Jordan structure at finite U. Curvature, unequal folded velocities, energy-dependent $M(\omega)$, nonscalar hybridization, or component-dependent interactions break (ii) and can shift or split the EP. (iii) The finite-volume continuation (twisted RTT relations, commuting transfer matrices, many-body Jordan blocks, and the four-state Fock lift, whose descendant *spin* roots meet at the projective point $\lambda = \infty$; no finite Anderson charge-root coalescence is asserted) is taken from Ref. [10] and is not rederived here. (iv) The interacting statements about the full-k pole EP are numerical: the embedding from exact finite-chain responses in App. C and the mean-field saddle of App. N; the residue audit of App. L tests a symmetric control.

Appendix N: Finite-U Kotliar–Ruckenstein saddle of the continuum bath

This appendix specifies the finite-U saddle of main Sec. III F and records its checks. It is a mean-field approximation to the interacting impurity, applied to the exact stationary bath of App. B 4. The state is the $T = 0$ grand-canonical ground state of the rotating-frame Hamiltonian at chemical potential μ, the same preparation assumption as App. C; the retarded $U = 0$ certificate does not need it. The parameters are those of Eq. (B20); $\bar{\Gamma}_0 = W^2/4 = 0.0025$ at $W = 0.10$.

1. Saddle

Local configuration probabilities (e, p_+, p_-, d) with $e + p_+ + p_- + d = 1$ are defined in a natural spin basis rotated by an angle θ in the real (x, z) plane, $\hat{n}_\theta = \sin\theta\,\sigma_x + \cos\theta\,\sigma_z$. With $n_\pm = p_\pm + d$,

$$r_\pm = \frac{\sqrt{e\,p_\pm} + \sqrt{p_\mp\,d}}{\sqrt{n_\pm(1-n_\pm)}}, \tag{N1}$$

$$R = \tfrac{1}{2}(r_+ + r_-)I + \tfrac{1}{2}(r_+ - r_-)\hat{n}_\theta, \qquad n = \tfrac{1}{2}(n_+ + n_-)I + \tfrac{1}{2}(n_+ - n_-)\hat{n}_\theta. \tag{N2}$$

A real symmetric field $\Lambda_f = \Lambda_{f0} I + \vec{\Lambda}_f \cdot \vec{\sigma}$, $\vec{\Lambda}_f = (\Lambda_{fx}, 0, \Lambda_{fz})$, enforces $n_f = n$, where

$$n_f = \tfrac{1}{2}I + \frac{1}{\pi}\int_0^\infty dy\,\mathrm{Herm}\,G_f(\mu + iy), \qquad G_f(z) = \big[zI - \Lambda_f - R\,\Sigma_d^{\mathrm{I}}(z)\,R\big]^{-1}, \tag{N3}$$

and Σ_d^{I} is Eq. (B17) with the static spin-even subtraction of App. B 4. The saddle is a stationary point of

$$\mathcal{L} = \Omega_{\mathrm{qp}}(R, \Lambda_f) + \mathrm{Tr}\,\big[(\epsilon_d I - \Lambda_f)\,n\big] + U\,d, \tag{N4}$$

where Ω_{qp} is the $T = 0$ grand potential of quasiparticles with propagator G_f. Stationarity in Λ_f gives $n_f = n$. Stationarity in the three independent probabilities and in θ gives the force equations, which contain the bath force

$$B = \frac{\partial\Omega_{\mathrm{qp}}}{\partial R} = \frac{1}{\pi}\int_0^\infty dy\,\mathrm{Re}\big[\Sigma_d^{\mathrm{I}} R G_f + G_f R\,\Sigma_d^{\mathrm{I}}\big]_{z=\mu+iy}. \tag{N5}$$

No paramagnetic, scalar-Z or Markov restriction is imposed. The θ equation is divided by the spin-density amplitude $(p_+ - p_-)/2$; this removes spurious unpolarized roots of the polar coordinate. The omitted complex spin direction is odd under complex conjugation of this real Hamiltonian and is tested separately for stability.

2. Resonance poles and EP equations

For complex z the saddle is held fixed and the bath kernel is continued to the outgoing sheet,

$$K_f^{\mathrm{II}}(z) = zI - \Lambda_f - R\,\Sigma_d^{\mathrm{II}}(z)\,R, \qquad F_{\mathrm{KR}} = \det K_f^{\mathrm{II}}, \qquad G_{d,\mathrm{coh}}^{\mathrm{II}} = R\,[K_f^{\mathrm{II}}]^{-1} R, \tag{N6}$$

with Σ_d^{II} of Eq. (B19) and both open channels continued. Since $\det K_f^{\mathrm{II}} = (\det R)^2 \det\big[R^{-1}(zI - \Lambda_f)R^{-1} - \Sigma_d^{\mathrm{II}}\big]$, the saddle acts as the linear self-energy

$$\Sigma_{\mathrm{KR}}(z) = (z - \epsilon_d)I - R^{-1}(zI - \Lambda_f)R^{-1} \tag{N7}$$

with Dyson convention $G_{d,\mathrm{coh}}^{-1} = (z - \epsilon_d)I - \Sigma_d - \Sigma_{\mathrm{KR}}$. In particular, $R = I$ and $\Lambda_f = \epsilon_d I + \Sigma_{\mathrm{HF}}$ give $\Sigma_{\mathrm{KR}} = \Sigma_{\mathrm{HF}}$. The EP equations $F_{\mathrm{KR}} = \partial_z F_{\mathrm{KR}} = 0$ are solved jointly with the saddle for (z, ϵ_d, V_O) at fixed W, or for (z, ϵ_d, W) at fixed V_O. At mean filling one, μ is an additional unknown with $\mathrm{Tr}\,n = 1$. The coherent weight of $G_{d,\mathrm{coh}}$ is $RR^\dagger$, not the full spectral sum, so local occupations are read from the probabilities. Certifying loops use the local-discriminant winding and pole tracking of App. B 4, with the saddle re-solved at every loop point.

3. Implementation and checks

Imaginary-axis integrals use Gauss–Legendre quadrature on each decade $10^{-11} \le y \le 10^{5}$ with analytic tails, and the radial integral is analytic.

Check	Result
Analytic radial integral against adaptive quadrature	1.5×10^{-16}
Density and force against a 194-dimensional Hermitian star	1.1×10^{-12} and 1.5×10^{-14}
Independent $\ln\det$ functional derivatives, including displaced states	discrepancy $\le 5.1 \times 10^{-10}$
98 accepted path records, orders 32 and 64	$\max\lvert\Delta R_{ij}\rvert < 5.2 \times 10^{-13}$; control/observable changes $< 1.4 \times 10^{-11}$
Held-state check at order 96	normalized residual $< 5.2 \times 10^{-12}$
Tail cutoff $10^{5} \to 10^{6}$ at three points	density change $< 2.3 \times 10^{-16}$; force change $< 4.4 \times 10^{-18}$
$U = 0$ limit	$Z_{\pm} = 1$; Eq. (B21) recovered; $\langle n_d \rangle = 0.03468$
First-order rates against Hartree–Fock (App. N 4)	agreement to 0.01–0.7%
Six perturbed seeds ($\langle n_d \rangle = 1$ EP)	same saddle
Local stability: $\mu = 0$ EPs at $U = 0.02,\ 0.2$; $\langle n_d \rangle = 1$ EP; occupied path	constrained and transverse curvatures positive

The 98 records include overlapping original and extended paths, including all 57 occupied-path points; failed continuation attempts are excluded. Density constraints are dimensionless, and force residuals are divided by the fixed energy scale 0.02. These finite-grid comparisons establish numerical convergence within the stated saddle equations, not an error bound relative to the exact interacting problem. Positive tested curvatures establish local stability, not a unique global minimum.

Four pole-exchange loops (the low-filling points at $U = 0.02$ and 0.2, the fixed-V_O point at $U = 0.0025$, and the filling-one EP) retain their winding and exchange on refinement from 128 to 256 intervals. At quadrature order 64, the largest pole residual is below 5.4×10^{-21} and the largest normalized saddle residual below 6.0×10^{-12}. Acceptance uses these residuals and pole tracking; solver stagnation flags are retained in the archived records. The tests are sampled numerical certificates.

At the low-filling EPs the two smallest constrained curvatures (2.8×10^{-9} and 3.8×10^{-7} at $U = 0.2$) lie along the logarithmic coordinates of the small probabilities d and p_{-}; the others are near 2×10^{-4}.

4. First-order limit

At $d = n_{+}n_{-}$ the factors are $r_{\pm} = 1$ with $\partial r_{\pm}/\partial d = 0$, so $R = I + O(U^2)$, and stationarity gives $\Lambda_f = \epsilon_d I + U[(\operatorname{Tr} n)I - n] + O(U^2)$: the Hartree–Fock field of main Sec. III F. An independent first-order calculation inserts this field, with the $U = 0$ continuum density matrix $\rho = \left(\begin{smallmatrix} 0.032677 & -0.0078 \\ -0.0078 & 0.002005 \end{smallmatrix}\right)$, into main Eq. (16). The saddle reproduces its rates:

Branch	Rates	KR saddle	Hartree–Fock
$\mu = 0$; (ϵ_d, V_O), W fixed	$d\epsilon_d/dU$, dV_O/dU	0.755, 30.41	0.761, 30.63
$\mu = 0$; (ϵ_d, W), V_O fixed	$d\epsilon_d/dU$, dW/dU	−0.552, −7.631	−0.551, −7.631
$\langle n_d \rangle = 1$; (ϵ_d, V_O, μ)	$d\epsilon_d/dU$, dV_O/dU, $d\mu/dU$	−2.625, −105.97, −2.233	−2.627, −105.99, −2.235

The first KR row is a finite difference over $\Delta U = 10^{-4}$; the second is a quadratic fit to $U \le 0.001$.

5. Low-filling branch and strong-U certificates

At $\mu = 0$ and $W = 0.10$, with (ϵ_d, V_O) retuned, every point below passes the joint residual $< 10^{-9}$, $\operatorname{Im} z < 0$ and the two-open-channel test; $|\vec{\Lambda}_f|$ is the traceless part of Λ_f.

U	$U/\bar{\Gamma}_0$	ϵ_d	V_O	$z_{\rm EP}$	$\langle n_d\rangle$	D_d	$1-Z_{\rm min}$	$\lvert\vec{\Lambda}_f\rvert$
0	0	0.015170	1.21342	$0.011973-0.002807i$	0.03468	4.7×10^{-6}	0	0
0.0005	0.2	0.015538	1.22820	$0.012370-0.002795i$	0.03324	4.1×10^{-6}	4.7×10^{-6}	8.1×10^{-6}
0.0025	1	0.016802	1.27873	$0.013731-0.002761i$	0.02907	2.6×10^{-6}	8.2×10^{-5}	3.3×10^{-5}
0.01	4	0.020009	1.40336	$0.017153-0.002702i$	0.02201	9.7×10^{-7}	5.6×10^{-4}	8.5×10^{-5}
0.02	8	0.022660	1.50252	$0.019953-0.002670i$	0.01833	4.6×10^{-7}	1.2×10^{-3}	1.2×10^{-4}
0.04	16	0.025828	1.61762	$0.023276-0.002644i$	0.01532	2.0×10^{-7}	2.1×10^{-3}	1.6×10^{-4}
0.08	32	0.028989	1.73072	$0.026572-0.002625i$	0.01320	7.6×10^{-8}	3.5×10^{-3}	1.9×10^{-4}
0.2	80	0.031930	1.83887	$0.029627-0.002610i$	0.01176	2.0×10^{-8}	5.8×10^{-3}	2.2×10^{-4}

The certifying loop is that of main Fig. 2 (half-widths 10^{-4} in ϵ_d and 10^{-2} in V_O, 128 points), with the saddle re-solved at every point:

Diagnostic	$U=0.02$	$U=0.2$
Local-discriminant winding	+1.000000000002	+1.000000000015
Endpoint pole permutation	(12)	(12)
Pole-separation half-winding	+0.500000000001	+0.500000000008
Minimum boundary modulus	9.69×10^{-10}	4.39×10^{-10}
Maximum phase increment	0.402	0.608
Minimum $\lvert\partial_z^2 F_{\rm KR}\rvert$	1.770	1.829
Maximum $\lvert\partial_z F_{\rm KR}\rvert$ at the critical points	1.1×10^{-15}	8.8×10^{-16}
Minimum pole separation	6.61×10^{-5}	4.38×10^{-5}
Maximum pole residual	2.6×10^{-21}	2.1×10^{-21}
Assignment/half-separation ratio	0.241	0.345
Maximum saddle residual on the loop	5.3×10^{-13}	4.6×10^{-12}
Kernel singular values at the EP	9.3×10^{-4}, 9×10^{-19}	7.7×10^{-4}, 4×10^{-18}
Threshold gap / pole half-width	2.78	4.15
Passivity margin $\Gamma_{\rm min}/\bar{\Gamma}$	0.815	0.847
$V_O/(\epsilon_O-\Omega)$	0.025	0.031

Both tracked poles keep two open channels on both loops. The double occupancy at $U=0.2$ is 0.09 of $\det n = 2.23\times10^{-7}$, while $\langle n_d\rangle = 0.0118$: the interaction is strongly repulsive but acts on a nearly empty level. The strong-U shift is kinematic. In the natural basis and the low-density limit, $r_\sigma^2 \simeq 1-n_{\bar\sigma}$, so the hybridization energy $K_{\bar\sigma}<0$ of one spin depends on the occupation of the other, and stationarity raises the level of spin σ by about $|K_{\bar\sigma}|$. A polarized bath makes $|K_\uparrow|\neq|K_\downarrow|$, so this shift becomes a spin field; it is the mean-field counterpart of Haldane's level renormalization [14].

6. Same-bath comparison

For the controls of App. C, (ϵ_d, W) at fixed $V_O = 1.21342201$, the KR saddle was solved on each of the four z-shifted 12-site chains, with the chain hybridization in Eqs. (N3) and (N5). The four saddles enter the same continuum two-cut kernel through $\langle R^{-2}\rangle z - \langle R^{-1}\Lambda_f R^{-1}\rangle$, with the saddles and EP iterated to self-consistency. This averages individual approximate self-energies. In contrast, the ED estimator is $\langle\Sigma_j G_j\rangle\langle G_j\rangle^{-1}$, obtained by averaging the equation-of-motion responses before taking their ratio. These prescriptions are generally unequal. An independent first-order check gives $dW/dU = -4.497758$ for the response-ratio average and -4.501190 for the average-density Hartree–Fock field. Thus the comparison retains a small estimator difference already at first order. The ED centers are those of App. C.

	W			ϵ_d				
U	ED	KR, chains	KR, cont.	ED	KR, chains	KR, cont.	shift ratio	$\frac{\text{KR,chains}}{\text{ED}}$
0.0010	0.095777	0.095672	0.092683	0.014840	0.014834	0.014634	1.025, 1.017	
0.0015	0.093859	0.093634	0.089258	0.014687	0.014675	0.014379	1.037, 1.026	
0.0020	0.092057	0.091678	0.085986	0.014543	0.014521	0.014135	1.048, 1.035	
0.0025	0.090363	0.089800	0.082865	0.014406	0.014373	0.013902	1.058, 1.043	

The last column gives the ratios of the shifts of W and ϵ_d. At $U = 0.0025$ the EPs are $0.011777 - 0.002258i$ (ED), $0.011776 - 0.002228i$ (KR, chains) and $0.011684 - 0.001874i$ (KR, continuum). Quadratic fits give initial slopes $dW/dU = -4.46$ (ED) and -4.49 (KR, chains), against -4.50 for Hartree–Fock with the z-averaged chain density matrix, and $d\epsilon_d/dU = -0.347$, -0.348 and -0.347. The approximate slope agreement does not remove the estimator difference described above. The continuum-to-chain ratio of the KR shifts is 1.68–1.69 in W and 1.59–1.60 in ϵ_d, close to the ratios of the impurity density matrices, 1.67 for the traceless part and 1.68 for the trace; the direction of $\vec{\rho}$ also differs by 5°. These ratios quantify this KR comparison, not the unknown exact continuum error. The certified point is the continuum KR solution in the table; transferring a chain method correction to it would require an additional, unvalidated assumption. On the interacting ellipse of App. C (half-widths 10^{-4} in ϵ_d and 10^{-3} in W, 128 points, saddle re-solved) the continuum KR EP gives winding -1, pole exchange (12) and half-winding $-1/2$, with maximum phase increment 0.144 and assignment ratio 0.237. With V_O fixed, the continuum branch continues to $U = 0.02$, where $W = 0.03104$, $\epsilon_d = 0.01054$, $z_{\rm EP} = 0.01021 - 0.00025i$ and $\langle n_d\rangle = 0.0039$; the next step, $U = 0.03$, did not converge and is not counted.

7. Mean filling one: fold of the EP branch

With $W = 0.10$ and μ, ϵ_d and V_O retuned for $\mathrm{Tr}\, n = 1$, the EP at $U = 0.0025$ lies at $z_{\rm EP} = 0.006535 - 0.002983i$, with $\epsilon_d = 0.008762$, $V_O = 0.94586$ and $\mu = 0.006657$; the bath filling therefore differs from that of the witness. Here $D_d = 0.229$ against $\det n = 0.241$, $P_1 = \langle n_d\rangle - 2D_d = 0.542$ and $Z_\pm = 0.9974$: a charge-fluctuating state, not a local moment. The kernel singular values are 1.54×10^{-3} and 2×10^{-18}. On the loop $\epsilon_d = \epsilon_{d,\rm EP} + 5 \times 10^{-6} \cos\phi$, $V_O = V_{O,\rm EP} + 10^{-3} \sin\phi$ with μ fixed and the saddle re-solved at 128 points, the poles exchange (error 8.5×10^{-16}) with half-winding $-1/2$ [main Fig. 7(a)]. The smallest constrained and transverse curvatures are 7.8×10^{-5} and 8.7×10^{-5}.

Treating U as an unknown and stepping V_O gives a local maximum $U_c = 0.0030826663 = 1.23\bar{\Gamma}_0$ at $V_O = 0.8566403$ and $\mu = 0.0044725258$, followed by a return to smaller U [main Fig. 7(b)]. Refinement at quadrature order 48 gives $dU/dV_O \simeq 0$, $d^2U/dV_O^2 \simeq -0.615$ and a nonsingular parameterized constraint Jacobian. This establishes a local fold of the exact-filling-one branch. With V_O fixed and (ϵ_d, W, μ) retuned, the stored continuation has a fold at $U_c = 0.007948$, $W = 0.1847$ [main Fig. 7(c)]. Both occur before a local moment develops.

The exact-filling constraint matters. Hold $\mu = 0.00447252582008$ at the first fold, keep $W = 0.10$, and solve the EP and saddle equations without imposing $\mathrm{Tr}\, n = 1$. At $U = 0.0032 > U_c$ a fresh order-64 solve gives

$$\epsilon_d = 0.006520501659, \qquad V_O = 0.845166571137,$$
$$z_{\rm EP} = 0.004515007629 - 0.003075182314i, \qquad \langle n_d\rangle = 0.98586922, \quad D_d = 0.21329561. \tag{N8}$$

Here $|F| = 1.75 \times 10^{-21}$, $|F_z| = 6.54 \times 10^{-16}$, $F_{zz} = 1.76239 - 0.37145i$, and the kernel singular values are 1.72×10^{-3} and 1.11×10^{-18}. The smallest constrained and transverse curvatures are positive, 9.56×10^{-5} and 1.20×10^{-4}. On a 128-point ellipse of radii 2×10^{-6} in ϵ_d and 2×10^{-4} in V_O, with the saddle re-solved, the discriminant winding is -1, the poles exchange, and the separation half-winding is $-1/2$. Both channels stay open. The maximum saddle and pole residuals are 1.63×10^{-15} and 4.98×10^{-21}. Thus the constrained fold does not exclude nearby EPs at higher U. The single-occupancy probability is only $P_1 = 0.55928$, so this is still a charge-fluctuating state, not a local-moment EP.

The separate occupied path in Fig. S5 fixes $\mu = 0.012$, $V_O = 1.21342201$, $W = 0.10$ and $\epsilon_d = 0.0151703 - U/2$; it is not particle–hole symmetric in this asymmetric bath.

U	$\langle n_d\rangle$	D_d	density eigenvalues	lower pole	upper resonance
0.02	0.97580	0.0598	0.0734, 0.9024	$0.003699 - 0.000306i$	$0.019904 - 0.002500i$
0.08	0.98677	0.0067	0.0072, 0.9796	-0.025655 (bound)	$0.049580 - 0.002411i$
0.20	0.99406	0.0013	0.0013, 0.9927	-0.085278 (bound)	$0.108818 - 0.002407i$

All three saddles are local minima, and the pole pairs have nonzero $\partial_z F_{\rm KR}$. The controls on this path do not enforce the EP equations, so the separated poles neither establish nor exclude an occupied-regime EP. The large-energy poles at strong U are coherent-saddle poles, not a controlled description of Hubbard satellites.

8. Scope

These are locally stable mean-field saddles. They do not bound the error of the KR approximation relative to the exact interacting impurity, prove a global free-energy minimum, or describe a screened Kondo state: in a polarized

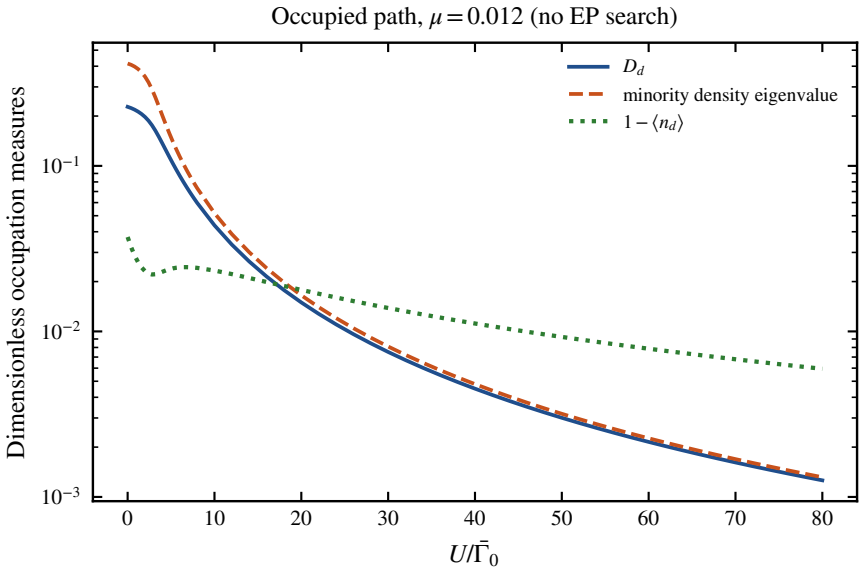


FIG. S5. **Separate occupied path in the continuum KR saddle** ($T = 0$). The controls are $\mu = 0.012$, $V_O = 1.21342201$, $W = 0.10$ and $\epsilon_d = 0.0151703 - U/2$, with fixed reference width $\bar{\Gamma}_0 = 0.0025$. Double occupancy D_d, the minority eigenvalue of the impurity density matrix, and $1 - \langle n_d \rangle$ decrease along this path. The bath is asymmetric, so the gate prescription does not impose particle–hole symmetry. The EP equations are not imposed: the split poles tabulated in App. N 7 neither establish nor exclude an occupied EP after retuning. The certified filling-one EP is shown in main Fig. 7.

bath the saddle gives a field-polarized static moment rather than a Kondo resonance with an exact width. The tabulated chain shifts differ by at most 6%, including a mesh-estimator difference; this is not an error bound for an exact continuum result. The low-filling strong-U branch remains nearly empty. The exactly filling-one folds concern two constrained control families and do not exclude the nearby EP of Eq. (N8). Testing a local-moment EP requires a controlled interacting continuum calculation, such as matrix-bath NRG with occupied-band convergence.

REPRODUCIBILITY

The accompanying archive contains the numerical scripts, parameter tables, and exported data used for the main and Supplemental figures. The public baseline is available in the version-controlled GitHub repository [15]. The original numerical baseline is Release v2.1.0, archived at doi:10.5281/zenodo.21888925; the all-versions record is doi:10.5281/zenodo.21434682. Release v2.1.0 includes the full-momentum certificate, SOC continuation, finite-U block-Wilson outputs, and finite-chain convergence gates reported here. Corrected real-frequency and Smith-delay figures and the direct filling checks are supplied in the accompanying revision archive; they are not contained in that earlier public snapshot. The dated 28 October revision in the repository supplies these additions; a matching fixed-version deposit will be identified after archiving. Its `checks_20261028/` directory contains the occupied-bath, two-sided pole, optimized-bath and exact PT-control scripts and outputs. The PT control is explicitly a different model and is not used to certify the driven-bath interaction claim. The accompanying source also includes `verify_finiteU_contact_algebra.py`, which reproduces the finite-U R-matrix, metric covariance, and YBE residuals quoted above. The script `full_k_two_cut_ep_certificate.py` reproduces the exact rotating-drive auxiliary closure, analytic full-momentum self-energy, two-cut continuation, nonlinear double-zero solve, local winding, and pole exchange of Fig. 2. The companion script `full_k_ep_soc_family.py` continues the double-zero locus in SOC, repeats the full topological certificate at three representative points, and generates main Fig. 3 and its tabulated data. The script `pt_phase_diagram.py` retains its historical filename but now evaluates the retarded upper rim and records positive half-width checks. With `make_pt_phase_figure.py` it reproduces the real-frequency splitting diagnostic in main Fig. 4, not a $\mathcal{PT}$ phase diagram. The certificate module is bundled beside it. The revision also supplies `make_smith_revision.py`, the matched filling outputs and a validation report distinguishing fresh reruns from archived evidence. The archived `finiteU_fullk_*` JSON summaries record the exact finite-chain interaction path, local winding, pole exchange, equation-of-motion identity, $N_z = 4 \to 8$ center check, and kept-space calibration of Sec. C. The archived `fullk_n_scaling_gate_summary.json` and `FULLK_N_SCALING_AUDIT.md` record the independently optimized $N = 10$ and 12 common-contour gate and Eq. (C3). The accompanying `archive_additions/redraw_main_figures.py` renders the

four main figures from the retained numerical records, including the finite-U summaries. It changes display coordinates and layout only; the pole values and topological checks are unchanged. The original `make_finiteU_fullk_exact_figure.py` is retained with its source data. The NRG archive additionally contains the independent transition-operator propagation, no-truncation regression, branch-tracking and convergence scripts, the gate summary, and the exported data used in Fig. S4. The revision archive also contains the finite-U Kotliar–Ruckenstein package with its JSON results (`matrix_saddle.py`, `filling_continuation.py` and `validate.py`); the strong-U certificates (`kr_certificate.py`); the same-chain comparison (`kr_on_chain.py` and `compare_bath.py`); the filling-one continuation and fold (`extend_n1.py` and `fold_n1.py`); the corrected $U \to \infty$ saddle with the closed form of Eq. (D9); and `make_fig5.py` and `make_filling_figures.py`, which render main Figs. 6 and 7, and Fig. S5, from the stored results.

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